

How to improve the market-based allocation mechanism of water rights for irrigation water users in China: a four-party game perspective

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Abstract

Promoting market-based trading in irrigation water rights and improving allocation mechanisms are critical for strengthening water governance and advancing sustainable agricultural development. This study develops a four-party evolutionary game model involving local government (LCG), irrigation district management unit (IMU), social capital (SCP), and irrigation water users (IWU) to examine the evolutionary dynamics and key determinants of the market-based allocation of irrigation water rights. The results show that: (1) The system admits 11 pure-strategy equilibria that are locally asymptotically stable under specific parameter conditions, and (1, 1, 1, 1) is the primary locally stable outcome under the baseline scenario. (2) System evolution is jointly shaped by the four actors' initial strategy probabilities, which mainly affect convergence speed and evolutionary trajectories rather than the long-run direction or final equilibrium. (3) Single-factor sensitivity analysis shows that higher service fee, expected return, reward-penalty intensity, repurchase price multiplier, subsidy rates for IMU and SCP, and reputational benefits, together with lower water resources taxes and a moderate reduction in reserved water volume, promote the market-based development of irrigation water-rights trading. (4) Two-factor sensitivity analysis identifies synergies between subsidy rates and reputational benefits, water resources taxes and reward-penalty intensity, expected return and management-technical factors, service fee and irrigation water-use benefits, and repurchase price multiplier and irrigation water-use benefits, with coordinated adjustments improving stability and accelerating convergence. This study provides theoretical and policy insights for improving market-based allocation of water resources and encouraging multi-actor participation in irrigation water-rights trading.

Introduction

Under the combined pressures of climate change and sustained economic growth, the global imbalance between water supply and demand is widening, and water scarcity is increasingly constraining sustainable socioeconomic development. Agriculture has long been the largest user of global freshwater, accounting for more than 70% of total freshwater withdrawals. However, irrigation practices remain relatively extensive, and management and metering systems remain underdeveloped, resulting in low output per unit of water, weak water-saving incentives, and persistent risks to food security and ecosystem stability (Qin et al. 2024). SDG 6 of the United Nations calls for improved water-use efficiency through integrated water resources management and for more equitable and

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sustainable water allocation. Target 6.4 further emphasises the need to improve water-use efficiency across sectors, reduce water stress, and strengthen sustainable water security (Anghileri et al. 2024; Vanham et al. 2018). Against this backdrop, agricultural water management is shifting from supply expansion towards efficiency-oriented demand-side management, which is central to achieving water security and supporting a sustainable agricultural transition. At the opening of the 2023 Rome Water Dialogue, the Food and Agriculture Organization of the United Nations (FAO) highlighted the need to adopt the 4 R principles of reducing, reusing, recycling, and replacing water use. FAO further emphasised that a systemic governance perspective is essential for enhancing water-use efficiency and resilience amid increasing global water-related risks and uncertainties. In China, per capita renewable freshwater resources are ~2400 m³, less than one quarter of the global average. Spatiotemporal disparities in water availability, together with low agricultural water-use efficiency, impose binding constraints on allocative efficiency and the realisation of water-saving potential. Accordingly, governance and allocation mechanisms that promote efficiency while safeguarding equity, together with stronger coordination between market instruments and institutional constraints, can help ease supply–demand tensions and reduce allocative distortions in water-scarce contexts. These mechanisms also provide an institutional basis for improving agricultural water-use efficiency and maintaining ecosystem stability.

Water-rights trading is a key institutional arrangement for introducing market mechanisms into water allocation. By enabling the transferability of water rights and transmitting price signals, it facilitates the orderly reallocation of water resources among actors, thereby improving allocative efficiency and supporting the transition towards market-based water governance. In China, water-rights trading has evolved from local experimentation into a more institutionalised and platform-based system. At the beginning of the twenty-first century, the interregional transfer between Dongyang and Yiwu in Zhejiang Province provided an early practical case of water-rights trading. Subsequent transactions in other regions increasingly adopted a hybrid model that combined government guidance with market participation, while trading arrangements gradually expanded to include interregional, cross-sectoral and user-to-user exchanges (Lv et al. 2021). As foundational measures such as water-rights registration, total water-use caps and trading-platform development have progressed, water-rights trading has moved from dispersed pilots into a more standardised operational phase (Moore, 2015; Jiang et al. 2022). Nevertheless, water-rights trading in China remains at a pivotal stage of transition from institution building to market deepening. Compared with mature water-rights markets in several developed countries, China's water-rights trading system still has a lower level of marketisation and operational efficiency. According to data from the China Water Rights Exchange, water-use rights trading in 2024 was recorded in 29 provincial-level regions, involving 12,997 transactions and a total traded volume of 1.499 billion m³. In terms of traded volume, interregional water-rights trades and water-withdrawal-rights trades dominated, together accounting for 93.6% of the total traded volume. By contrast, trades involving irrigation water users accounted for 88.4% of all transactions, yet their traded volume was only 0.182 billion m³, representing less than 7% of the total traded volume. This “high-count, low-volume” pattern suggests that trading among irrigation water users has not yet developed a scaled-up and consolidated operating mechanism, leaving substantial scope for further gains in allocative efficiency (Sheng et al. 2024).

In terms of theory, the market-based allocation problem in irrigation water-rights trading falls within the domains of common-resource governance and institutional economics (Schlager and Ostrom, 1992; Meinzen-Dick, 2007). Coasean theory suggests that, when property rights are well defined and transaction costs are negligible, parties can achieve efficient allocation through voluntary bargaining (Friesen et al. 2023). However, water resources have strong public-good attributes and externalities, and market coordination is often constrained in practice by unclear rights delineation and high transaction costs (Ehrman, 2020). Property-rights theory further argues that clear rights delineation is a prerequisite for shifting common resources from administrative allocation to market-based allocation and provides the basis for designing incentive and constraint mechanisms (Sikor et al. 2017). Although water rights are divisible and transferable, they also involve public responsibilities and ecological sustainability requirements. Accordingly, a well-defined property-rights regime delineates rights boundaries and establishes a framework of responsibility and accountability, thereby providing institutional support for the effective operation of incentive mechanisms. From a behavioural perspective, incentive-design theory posits that aligning benefits with responsibilities can strengthen intrinsic motivation for water saving and sustainable use under a regime that combines incentives and constraints, thereby mitigating the tragedy of the commons (Ostrom, 1990; Farley and Costanza,

2010). The effectiveness of behavioural incentives depends on the structure of interactions among multiple actors. Common-resource governance theory extends the analytical lens to multi-actor interactions and emphasises the integration of institutionalised cooperation with the strategic relations among actors. By embedding incentives, constraints and coordination mechanisms into institutional design, such a governance framework can dynamically balance stakeholder interests while improving allocative efficiency and overall governance performance (Ostrom, 2009; Pahl-Wostl, 2009).

Existing research on the market-based development of irrigation water-rights trading can be broadly grouped into three main strands. In terms of research objects, property-rights institutions are widely regarded as the institutional foundation of market-based water allocation. However, cross-country differences in institutional arrangements have led to considerable heterogeneity in the legal attributes of water rights and the corresponding allocation logic. In countries such as the United States, Australia and Chile, water is often treated as a scarce resource with property-like attributes. Under privatised or divisible rights arrangements, these systems allow investors, irrigation associations and private firms to participate in water allocation and typically operate through a combination of public regulation and market coordination (Grafton et al. 2011; Del Moral and Saurí, 1999). By contrast, China's water-rights regime is embedded in a framework of public ownership and administratively guided allocation. Water resources are state-owned, and water rights are mainly expressed through abstraction permits or use quotas, leaving the legal basis of trading and the boundaries of rights in need of further institutional clarification. Under this arrangement, water allocation remains largely government-led, while private participation and market coordination remain comparatively constrained (Deng et al. 2023; Wang et al. 2025). Taken together, these studies provide an important institutional basis for understanding the market-based development of irrigation water-rights trading across different governance contexts.

From an analytical perspective, existing studies on irrigation water-rights trading have largely relied on comparative-static approaches, focusing on outcome-oriented issues such as pricing mechanisms, behavioural responses to water-saving incentives, water-use efficiency and economic performance. Studies on pricing mechanisms suggest that price formation not only reflects water scarcity but also guides the reallocation of water among users through price signals, thereby helping to reconcile ecological constraints with economic objectives (Wheeler et al. 2025). Behavioural studies commonly adopt a "property-rights incentives and price constraints" perspective, arguing that tradability enables water-saving surpluses to be monetised, while price signals reshape water-use behaviour by strengthening opportunity-cost considerations (Chen et al. 2023; Gao et al. 2020; Wheeler et al. 2013; Hadjigeorgalis, 2008). Studies of efficiency and performance further indicate that trading-induced reallocation can redirect water towards users with higher marginal returns, thereby improving overall allocative efficiency (Debaere and Li, 2020). In addition, studies of institutional performance have examined the effects of irrigation water-rights trading on agricultural costs, farm income and regional economic activity, while emphasising the need for multidimensional performance assessment (Deng and Zhang, 2025; Huang et al. 2017). This body of literature provides substantial evidence on the economic and policy effects of irrigation water-rights trading.

With respect to determinants, existing studies can be broadly grouped into two strands: single-factor analyses and multi-factor interaction analyses. Single-factor studies typically examine one dimension at a time, such as technical conditions, resource endowments, or institutional arrangements, and assess the marginal effects of water-saving technologies, cropping patterns, and irrigation-related regulations on the development of water-rights trading (Pfeiffer and Lin, 2014; Zeleke and Luckett, 2025; Wheeler et al. 2017). As research has progressed, increasing attention has been paid to interactions among multiple determinants. Using interprovincial or basin-level panel data, recent studies have employed multi-period difference-in-differences models, panel regressions and spatial econometric methods, often combined with mediation analysis, to examine how transaction costs, market power, price elasticity, regulatory intensity and intermediary services shape water-rights trading outcomes (Zhao et al. 2024; Du et al. 2022). These studies suggest that the market-based development of irrigation water-rights trading is not driven by any single factor but is shaped by the combined influence of institutional, technical and economic conditions. This strand of literature highlights the complexity of the determinants underlying irrigation water-rights trading and provides an important foundation for further analysing their joint effects.

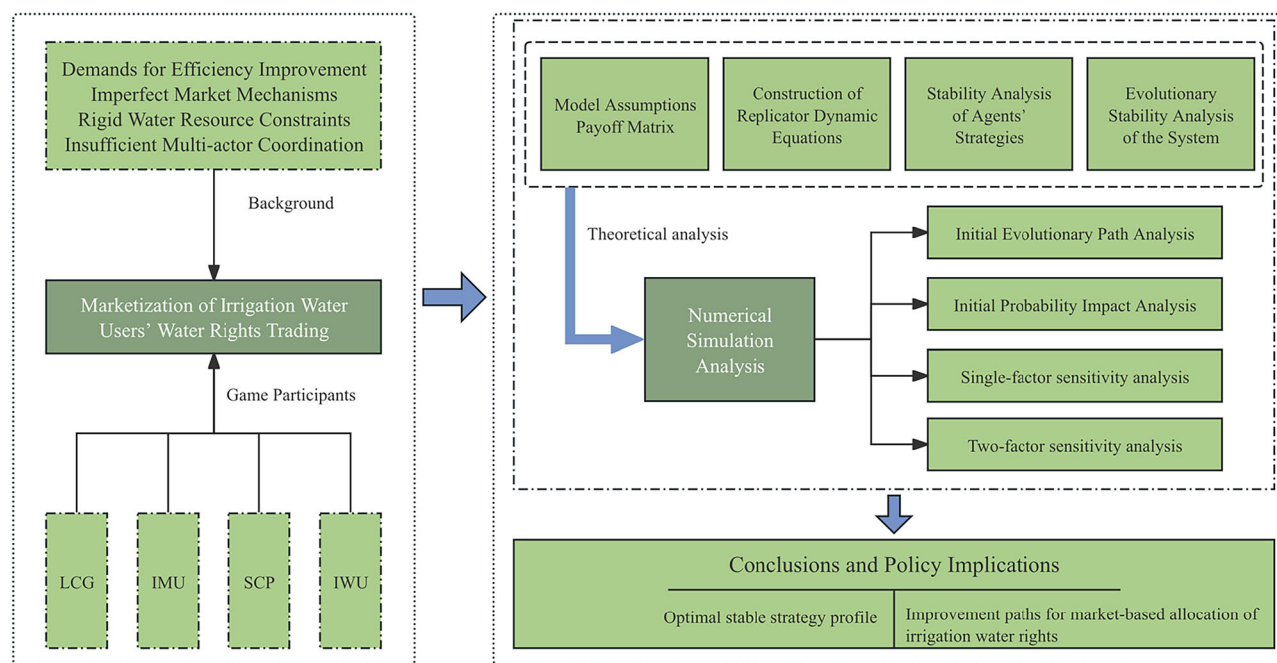


Fig. 1 Overall research framework.

Despite these advances, the literature on irrigation water-rights trading and its market-based development still has several limitations. In terms of research objects, existing studies have mainly examined interactions among local government (LCG), irrigation district management unit (IMU) and irrigation water users (IWU), while paying relatively limited attention to the role of social capital (SCP) in the market-based allocation of irrigation water rights, particularly in the context of emerging public-private governance arrangements in China. From an analytical perspective, existing studies have largely focused on outcome-oriented issues, such as pricing mechanisms, water-saving effects, water-use efficiency and economic performance, whereas dynamic strategic interactions and evolutionary mechanisms among multiple stakeholders remain insufficiently explored. Regarding the determinants of market development, although existing studies have identified a range of institutional, technical and economic factors, systematic evidence remains limited on how heterogeneous factors and their interactions jointly shape the evolutionary trajectories, convergence processes and stable outcomes of irrigation water-rights trading.

To address these limitations, this study makes three main contributions. First, we construct a four-party evolutionary game model that integrates LCG, IMU, SCP, and IWU within a unified analytical framework. By explicitly introducing SCP, this study extends the actor structure considered in previous research and better reflects the institutional complexity of irrigation water-rights trading in China. Second, we move beyond static outcome assessment and examine the dynamic strategic interactions underlying irrigation water-rights trading, thereby revealing the adjustment processes, equilibrium conditions and evolutionary mechanisms of market-based allocation among multiple stakeholders. Third, we develop an integrated parameter framework covering foundational, policy-related, and management-technical factors, and systematically examine both single-factor and two-factor effects on the market-based development process. In doing so, this study identifies the directional effects of key determinants and clarifies how their interactions influence convergence speed, evolutionary paths and stable equilibria.

The remainder of this study is structured as follows (Fig. 1). Section 2 develops a four-party evolutionary game model involving LCG, IMU, SCP, and IWU and characterises the strategic interactions among these actors in water-rights trading. Section 3 analyses the evolutionary dynamics and stability of the model and identifies patterns associated with different strategy profiles. Section 4 uses numerical simulations to examine the evolutionary trajectories of actors' strategies and their key determinants and to investigate the mechanisms underpinning stable outcomes. Section 5 provides a comprehensive discussion and offers further interpretation and extension of the

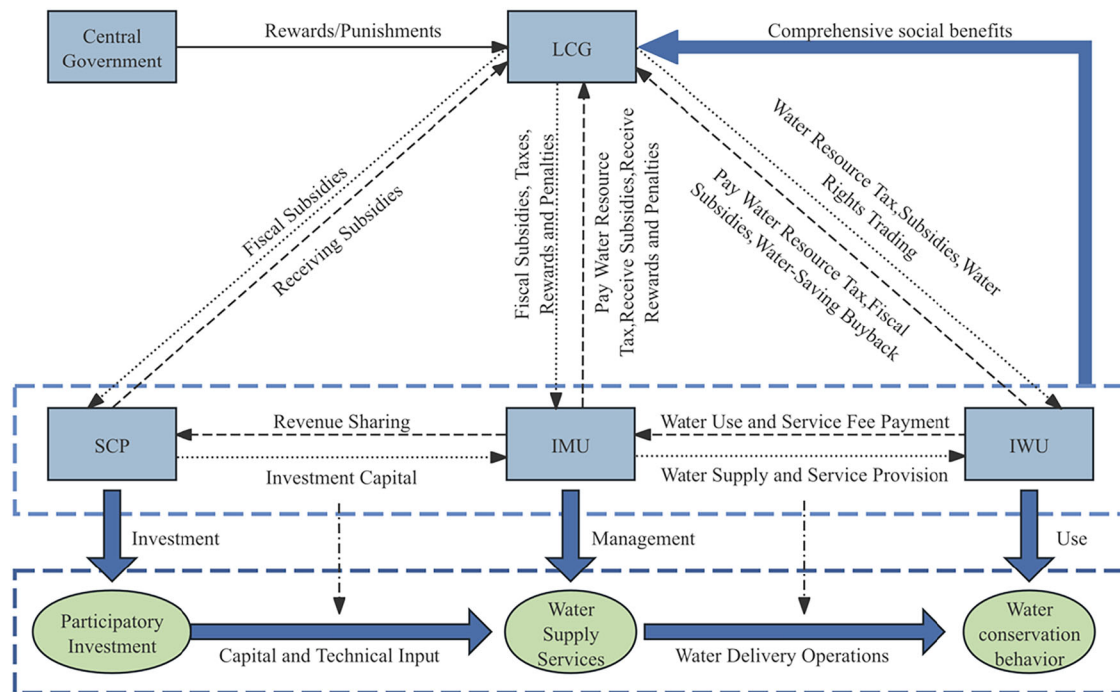


Fig. 2 Logical framework of evolutionary game interactions in water-rights trading among irrigation water users.

results. Section 6 summarises the main findings and outlines policy implications for improving the market-based allocation of irrigation water rights.

Methods

Problem description

With a large population and substantial food demand, China relies on the rational allocation and efficient use of water resources to safeguard food security and advance agricultural modernisation. Official statistics indicate that China's surface water resources total 2989.56 billion m³, while its groundwater resources total 867.92 billion m³. The spatial distribution of these resources remains highly uneven, with relative abundance in the southeast and scarcity in the northwest. In China's water-withdrawal structure, agriculture accounts for 61.6% of total withdrawals, with irrigation representing the largest share of agricultural water use. However, conventional irrigation remains widespread, and overall agricultural water-use efficiency remains low, resulting in substantial water losses (Levidow et al. 2014; Sajjad et al. 2025). To improve agricultural water-use efficiency and support sustainable water use, water-use rights reform has become a core institutional instrument (Xu and Yang, 2025; Xu et al. 2023). Against this backdrop, relevant authorities jointly issued a policy document on advancing water-use rights reform, setting out a market-oriented pathway. The document calls for the initial allocation system to be largely established by 2025 and for trading arrangements for irrigation water users to be further improved.

The market-based allocation of irrigation water rights among irrigation water users operates as a complex system characterised by multi-actor participation, strategic interactions, and co-evolutionary dynamics, involving LCG, IMU, SCP, and IWU. LCG chooses between an incentive-oriented strategy and direct intervention. It uses policy instruments such as water pricing, subsidies, and reward-penalty intensity to influence the decisions of the other actors. IMU chooses between high-efficiency management and inert management. This choice directly affects water-delivery efficiency and the structure of operating costs. SCP chooses between active participation and opportunistic

participation by weighing expected return against investment costs. Its level of engagement affects infrastructure quality and operational performance. IWU chooses between active and passive water-saving behaviour. This decision is jointly shaped by government incentives, management efficiency, and SCP investment. Given the strong institutional constraints and the complexity of multi-actor coordination in the market-based trading of irrigation water rights, we develop an evolutionary game model incorporating the four actors described above (Fig. 2). We use this model to examine strategic dynamics and equilibrium outcomes by tracing the evolutionary trajectories of the actors and deriving the corresponding stability conditions. The results provide a theoretical basis for optimising market-based allocation mechanisms for irrigation water rights.

Basic assumptions of the model

Litong District in Wuzhong City, Ningxia Hui Autonomous Region, was selected as the representative case for this study. Using this case as the empirical background, the model hypotheses and parameter settings were developed based on the institutional context, case evidence and field investigations. Policy documents related to water-rights reform, agricultural water pricing, water resources tax and water-saving incentives provide the institutional basis for the model. Project materials from the Modernised Ecological Irrigation District Project in Litong District, together with fieldwork and semi-structured interviews with relevant stakeholders, provide empirical support for scenario construction and parameter calibration. In the subsequent sensitivity analysis, selected key parameters were extended within reasonable ranges to capture their potential effects under different practical conditions. The main hypotheses are presented below.

Hypothesis 1. The model includes four stakeholder groups, namely LCG, IMU, SCP, and IWU. Each group is represented as a population of agents with similar institutional roles rather than as a single fully rational decision-maker. Under bounded rationality, agents within each population gradually adjust their strategy choices through learning, imitation and payoff feedback, so that the shares of alternative strategies evolve over time towards relatively stable states.

Hypothesis 2. Let x denote the probability that LCG adopts an incentive-oriented strategy, and let $1 - x$ denote the probability that it adopts direct intervention. Let y denote the probability that IMU adopts a high-efficiency management strategy, and let $1 - y$ denote the probability that it adopts an inert management strategy. Let z denote the probability that SCP adopts an active participation strategy, and let $1 - z$ denote the probability that it adopts an opportunistic participation strategy. Let w denote the probability that IWU adopts an active water-saving strategy, and let $1 - w$ denote the probability that it adopts a passive water-saving strategy. It is assumed that $0 < x, y, z, w < 1$.

Hypothesis 3. LCG applies a quota-based pricing scheme to agricultural water use. The quota volume, quota price and over-quota price are denoted by W_0 , P_0 , and P_1 , respectively. For the water saved by IWU, LCG buys back the saved volume at nP_0 . With respect to aggregate social welfare, if IMU, SCP and IWU all choose cooperative strategies, LCG receives a reward r_s from the central government. If all three choose non-cooperative strategies, LCG incurs an equivalent penalty r_s . LCG incurs a base administrative cost C_b to implement its management policy. Under the direct intervention strategy, LCG additionally incurs a maintenance cost C_f and imposes a fixed penalty f on IMU when IMU adopts an inert management strategy. Under the incentive-oriented strategy, LCG provides IMU with a reward f when IMU adopts a high-efficiency management strategy. LCG further subsidises the fixed costs of IMU and SCP, as well as the cost incurred by IWU when it adopts an active water-saving strategy. The corresponding subsidy rates are λ_1 , λ_2 , and λ_3 , respectively. LCG also derives reputational benefits from providing these subsidies, captured by the coefficient L .

Hypothesis 4. In the practical operation of irrigation water-rights trading, IMU is responsible for water delivery, infrastructure maintenance, operational management, and service coordination within the irrigation district. Accordingly, under the high-efficiency management strategy, IMU invests in infrastructure and irrigation-district maintenance. The fixed cost, unit water-supply cost coefficient, and reserved water volume coefficient set by IMU based on local supply conditions are denoted by C_a , θ , and α , respectively. Under the inert management strategy, IMU's water utilisation coefficient is η . IMU charges IWU a service fee P_f . The fee is waived when IWU adopts an

Table 1 Definitions of parameters.

Parameter	Definition	Range	Parameter	Definition	Range
x	Probability of LCG choosing the incentive-oriented strategy	$x \in [0, 1]$	λ_3	The subsidy rate for IWU's irrigation-related cost under LCG's incentive-oriented strategy	$\lambda_3 > 0$
y	Probability of IMU adopting the high-efficiency management strategy	$y \in [0, 1]$	L	Reputational benefit coefficient of LCG under the incentive-oriented strategy	$L > 1$
z	Probability of SCP adopting the active participation strategy	$z \in [0, 1]$	C_a	Fixed cost of IMU under the high-efficiency management strategy	$C_a > 0$
w	Probability of IWU adopting the active water-saving strategy	$w \in [0, 1]$	η	Water utilisation coefficient of IMU under the inert management strategy	$\eta \in [0, 1]$
W_0	The quota volume for agricultural water use	$W_0 > 0$	α	Reserved water volume coefficient of IMU based on current water supply conditions	$\alpha \in [0, 1]$
W_1	Water consumption of IWU under an active water-saving strategy	$W_1 > 0$	θ	Unit water supply cost coefficient of IMU under the high-efficiency management strategy	$\theta > 0$
W_2	Water consumption of IWU under a passive water-saving strategy	$W_2 > 0$	P_f	Service fee charged by IMU under the high-efficiency management strategy	$P_f > 0$
P_0	Quota price for irrigation water	$P_0 > 0$	V_f	Revenue share to IMU under the joint adoption of the active participation strategy by SCP and the high-efficiency management strategy by IMU	$V_f > 0$
P_1	The over-quota irrigation water price	$P_1 > 0$	V_e	Revenue share to IMU under the joint adoption of the opportunistic participation strategy by IMU by SCP and the high-efficiency management strategy by IMU	$V_e > 0$
C_b	The base administrative cost incurred by LCG	$C_b > 0$	t_1	Water resources tax paid by IMU to the LCG	$t_1 > 0$
n	Repurchase price multiplier for water rights by LCG	$n > 0$	t_2	Water resources tax paid by IWU to the LCG	$t_2 > 0$
r_s	The central government's reward and penalty for LCG based on aggregate social welfare	$r_s > 0$	C_0	Total fixed cost incurred by SCP under the active participation strategy	$C_0 > 0$
C_f	Maintenance cost of the irrigation district	$C_f > 0$	C_t	Unit variable cost incurred by SCP under the active participation strategy	$C_t > 0$
f	Reward-penalty intensity imposed by LCG on IMU	$f > 0$	r_u	Expected return of SCP under the active participation strategy	$r_u > 0$
λ_1	The subsidy rate for IMU's fixed cost under the LCG's incentive-oriented strategy	$\lambda_1 > 0$	u	Irrigation water-use benefits obtained by IWU	$u > 0$
λ_2	The subsidy rate for SCP's fixed cost under LCG's incentive-oriented strategy	$\lambda_2 > 0$	C_i	Irrigation-related costs incurred by IWU	$C_i > 0$

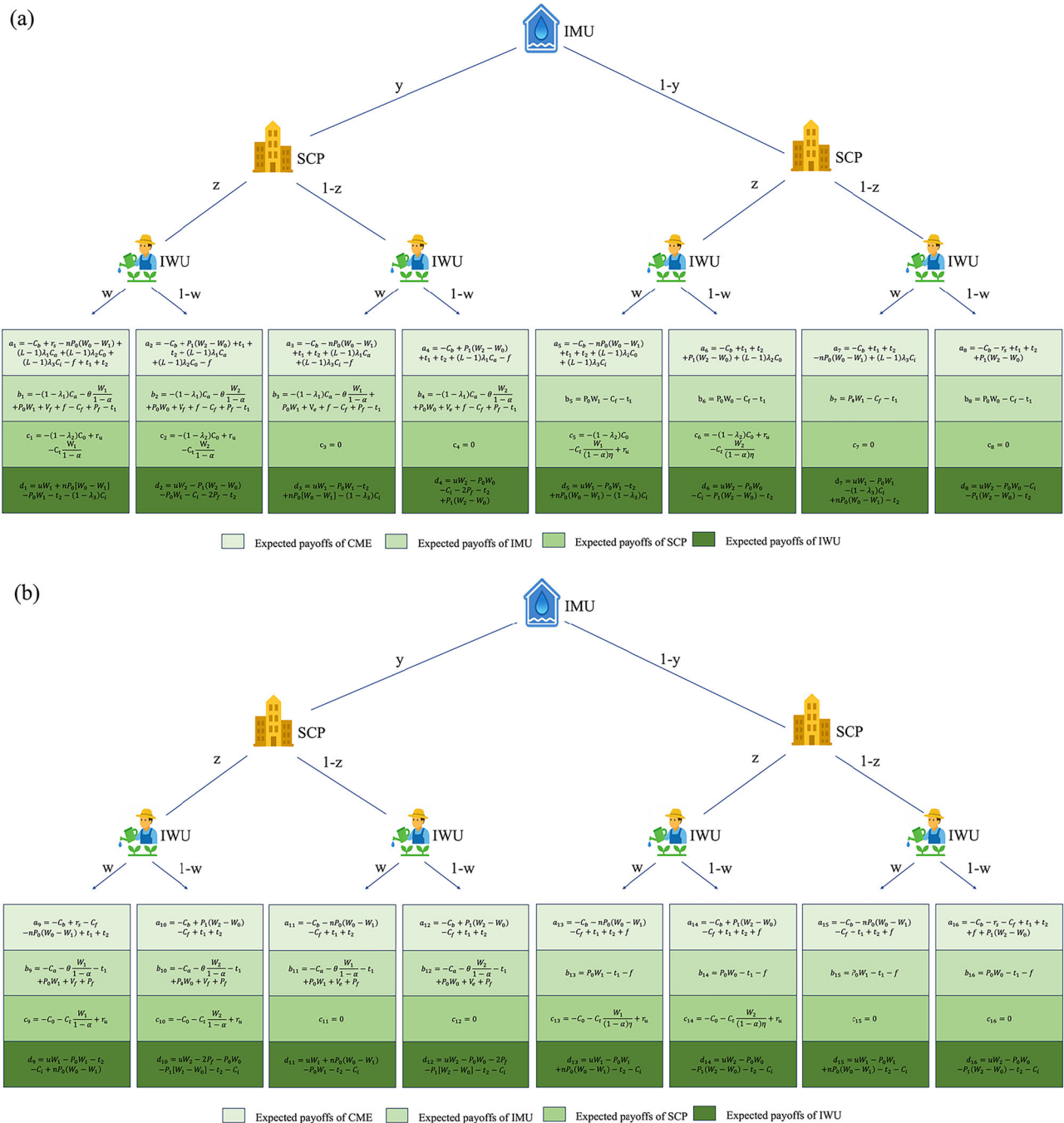


Fig. 3 Payoff matrices under the two strategies adopted by LCG: **a** incentive-oriented strategy; **b** direct intervention strategy.

active water-saving strategy, whereas IWU pays $2P_f$ under a passive water-saving strategy. When LCG adopts the incentive-oriented strategy, IMU bears the maintenance cost C_f . In its collaboration with SCP, IMU receives a revenue share V_f when SCP adopts an active participation strategy and V_e when SCP adopts an opportunistic participation strategy. IMU also pays a water resources tax t_1 to LCG.

Hypothesis 5. When SCP adopts an active participation strategy, it incurs a total fixed cost C_0 and a unit variable cost C_t , and earns an expected return r_u from the invested water volume.

Hypothesis 6. IWU's irrigation water use differs across water-saving strategies. Under the active water-saving strategy, IWU adopts water-saving irrigation practices and consumes W_1 units of water. Under the passive water-

saving strategy, IWU maintains conventional irrigation practices and consumes W_2 units of water, where $W_1 < W_2$. IWU derives water-use benefits u from irrigation and incurs an irrigation-related cost C_i . Under direct intervention by LCG, IWU pays the water resources tax t_2 . Table 1 summarises the parameter definitions and notation used in the model.

Based on the above assumptions and parameter settings, Fig. 3 presents the payoff matrices for each actor when LCG adopts the incentive-oriented strategy and the direct intervention strategy, respectively.

Results of evolutionarily stable strategies

Evolutionary game theory provides a suitable framework for analysing strategic interactions under bounded rationality (Taylor and Jonker, 1978; Na et al. 2024). This study employs replicator dynamics to characterise the payoff-driven adaptive evolution of LCG, IMU, SCP, and IWU in the context of irrigation water-rights trading. Compared with best-response dynamics, stochastic learning, and hierarchical adjustment mechanisms, replicator dynamics more effectively capture the payoff-comparison logic of boundedly rational actors in repeated interactions and provide a formal basis for analysing equilibrium selection and local stability. Accordingly, this section develops the replicator dynamic system and examines the local stability of the corresponding equilibrium points.

Construction of replicator dynamic equations

Analysis of evolutionarily stable strategies of LCG

The expected payoffs of LCG under the direct intervention strategy and the incentive-oriented strategy are denoted by U_{G_1} and U_{G_2} , respectively, and the corresponding average expected payoff is denoted by U_G . Accordingly, the replicator dynamic equation governing the evolution of LCG's strategic behaviour, together with its first-order derivative, is presented in Eqs. (2) and (3).

$$\begin{aligned}
 U_{G_1} = & yzwa_1 + yz(1-w)a_2 \\
 & + y(1-z)wa_3 + y(1-z)(1-w)a_4 \\
 & + (1-y)zwa_5 + (1-y)z(1-w)a_6 \\
 & + (1-y)(1-z)wa_7 + (1-y)(1-z)(1-w)a_8 \\
 U_{G_2} = & yzwa_9 + yz(1-w)a_{10} \\
 & + y(1-z)wa_{11} + y(1-z)(1-w)a_{12} \\
 & + (1-y)zwa_{13} + (1-y)z(1-w)a_{14} \\
 & + (1-y)(1-z)wa_{15} + (1-y)(1-z)(1-w)a_{16} \\
 U_G = & xU_{G_1} + (1-x)U_{G_2}
 \end{aligned} \tag{1}$$

$$F(x) = x(U_{G_1} - U_G) = x(1-x)(U_{G_1} - U_{G_2}) = x(1-x)A(y, z, w) \tag{2}$$

$$\begin{aligned}
 A(y, z, w) = & -f + C_f - t_1 - t_2 \\
 & + y(L-1)C_a\lambda_1 \\
 & + z(L-1)C_0\lambda_2 \\
 & + w(L-1)C_i\lambda_3 \\
 F'(x) = & (1-2x)A(y, z, w)
 \end{aligned} \tag{3}$$

Accordingly, LCG's choice between the incentive-oriented strategy and direct intervention strategy is jointly shaped by the strategy selection probabilities of the other three actors, subsidy intensity, reputational benefits, reward-penalty intensity, irrigation district maintenance costs, and water resources taxes. According to the stability conditions of the replicator dynamic system, LCG's strategic probability reaches a stable equilibrium when $F(x) = 0$ and $F'(x) < 0$.

Proposition 1. When $z < z_1$, LCG adopts the direct intervention strategy as the stable outcome. When $z > z_1$, the stable outcome shifts to the incentive-oriented strategy. When $z = z_1$, LCG's stable strategy is indeterminate. The threshold value is defined as $z_1 = \frac{f - C_f + t_1 + t_2 - y(L-1)C_a\lambda_1 - w(L-1)C_i\lambda_3}{(L-1)C_0\lambda_2}$.

Proposition 1 indicates that LCG's strategic choice is partly shaped by SCP's investment willingness. A higher level of investment willingness increases LCG's propensity to adopt a more flexible incentive-oriented strategy. Conversely, when SCP's investment willingness is weak, LCG tends to rely more heavily on a direct intervention strategy primarily characterised by administrative intervention and regulatory enforcement. The threshold in Proposition 1 captures the minimum level of SCP participation required for LCG to shift from direct intervention to incentive-oriented strategy. In practical terms, when SCP participation remains below this critical level, the market mechanism lacks sufficient external support, making administrative control the more likely policy choice. Once this threshold is exceeded, incentive-based governance becomes more feasible and institutionally attractive.

Based on Proposition 1, the phase diagram of LCG's strategic choice is presented in Appendix A, Fig. A1.

Corollary 1.1. When LCG's reputational benefits are high, irrigation district maintenance costs are relatively high, and subsidy rates are elevated, LCG is more likely to adopt an incentive-oriented strategy. By contrast, when the reward-penalty intensity and the level of water resources taxes are high, LCG tends to favour a direct intervention strategy.

Corollary 1.2. When LCG's reputational benefits L exceed the threshold L^* , defined as $L^* = 1 + \frac{f - C_f + t_1 + t_2}{yC_a\lambda_1 + zC_0\lambda_2 + wC_i\lambda_3}$, LCG is more likely to adopt the incentive-oriented strategy. In the presence of stronger reward-penalty intensity or rising water resources tax levels, LCG must further enhance its reputational benefits to sustain a stable preference for this strategy. By contrast, when irrigation district maintenance costs and subsidy rates increase and other actors exhibit active participation, the threshold L^* declines, thereby increasing the likelihood that LCG will adopt the incentive-oriented strategy.

Analysis of evolutionarily stable strategies of IMU

The expected payoffs of IMU under the high-efficiency management strategy and the inert management strategy are denoted by U_{I_1} and U_{I_2} , respectively, with the average expected payoff denoted by U_I . Accordingly, the replicator dynamic equation governing the evolution of IMU's strategic behaviour, together with its first-order derivative, is

presented in Eqs. (5) and (6).

$$\begin{aligned}
 U_{I_1} &= xzw b_1 + xz(1-w)b_2 \\
 &\quad + x(1-z)wb_3 + x(1-z)wb_4 \\
 &\quad + (1-x)zwb_9 + (1-x)z(1-w)b_{10} \\
 &\quad + (1-x)(1-z)wb_{11} + (1-x)(1-z)(1-w)b_{12} \\
 U_{I_2} &= xzw b_5 + xz(1-w)b_6 \\
 &\quad + x(1-z)wb_7 + x(1-z)(1-w)b_8 \\
 &\quad + (1-x)zwb_{13} + (1-x)z(1-w)b_{14} \\
 &\quad + (1-x)(1-z)wb_{15} + (1-x)(1-z)(1-w)b_{16} \\
 U_I &= yU_{I_1} + (1-y)U_{I_2}
 \end{aligned} \tag{4}$$

$$\begin{aligned}
 F(y) &= y(U_{I_1} - U_I) = y(1-y)(U_{I_1} - U_{I_2}) \\
 &= y(1-y)B(x, z, w)
 \end{aligned} \tag{5}$$

$$\begin{aligned}
 B(x, z, w) &= -C_a + P_f + V_e + f - \frac{W_2\theta}{1-\alpha} \\
 &\quad + xC_a\lambda_1 \\
 &\quad + z(V_f - V_e) \\
 &\quad + w\frac{\theta(W_2 - W_1)}{1-\alpha} \\
 F'(y) &= (1-2y)B(x, z, w)
 \end{aligned} \tag{6}$$

Accordingly, IMU's decision to adopt the high-efficiency management strategy is jointly influenced by the strategy selection probabilities of the other three actors, subsidy levels, revenues, reserved water volume, reward-penalty intensity, and service fee. According to the stability conditions of the replicator dynamic system, IMU's strategic probability attains a stable equilibrium when $F(y) = 0$ and $F'(y) < 0$.

Proposition 2. When $z < z_2$, IMU adopts the inert management strategy as the stable outcome. When $z > z_2$, the stable outcome shifts to the high-efficiency management strategy. When $z = z_2$, IMU's stable strategy is indeterminate. The threshold value is defined as $z_2 = \frac{C_a - P_f - V_e - f + \frac{W_2\theta}{1-\alpha} - xC_a\lambda_1 - w\frac{\theta(W_2 - W_1)}{1-\alpha}}{(V_f - V_e)}$.

Proposition 2 indicates that IMU's strategic choice is partly shaped by SCP's willingness to invest. When SCP's willingness to invest is low, the high-efficiency management strategy imposes relatively high cost pressure on IMU, increasing its likelihood of shifting to the inert management strategy. As SCP's willingness to invest strengthens and external support capacity improves, IMU becomes more likely to adopt the high-efficiency management strategy. The threshold in Proposition 2 represents the minimum support condition under which IMU can economically sustain high-efficiency management. This implies that management improvement depends not only on IMU's own strategic willingness but also on whether government incentives, service fee, and cooperation with SCP can jointly offset the additional costs associated with efficient operation.

Based on Proposition 2, the phase diagram of IMU's strategic choice is presented in Appendix A, Fig. A2.

Corollary 2.1. IMU is more likely to adopt the high-efficiency management strategy when LCG provides higher subsidies, when reward-penalty intensity increases or when irrigation service fee rise. In contrast, a larger reserved water volume increases the likelihood that IMU adopts an inert management strategy.

Corollary 2.2. When the subsidy provided by LCG to IMU, λ_1 , exceeds the threshold λ_1^* , defined as $\lambda_1^* = -\frac{C_a - P_f - V_e - f + \frac{W_2\theta}{1-\alpha} + z(V_f - V_e) + w\frac{\theta(W_2 - W_1)}{1-\alpha}}{xC_a}$, IMU is more likely to adopt the high-efficiency management strategy. If the reserved water volume coefficient is relatively large, a higher subsidy is required to sustain this strategic choice. Conversely, increases in the service fee level or the reward-penalty intensity reduce the threshold λ_1^* , thereby increasing the likelihood that IMU will adopt the high-efficiency management strategy.

Analysis of evolutionarily stable strategies of SCP

The expected payoffs of SCP under the active participation strategy and the opportunistic participation strategy are denoted by U_{S_1} and U_{S_2} , respectively, with the average expected payoff given by U_S . The replicator dynamics equation governing SCP's behavioural strategy and its first-order derivative are presented in Eqs. (8) and (9).

$$\begin{aligned}
 U_{S_1} = & xywc_1 + xy(1-w)c_2 \\
 & + x(1-y)wc_5 + x(1-y)(1-w)c_6 \\
 & + (1-x)ywC_9 + (1-x)y(1-w)C_{10} \\
 & + (1-x)(1-y)wC_{13} + (1-x)(1-y)(1-w)C_{14} \\
 U_{S_2} = & xywC_3 + xy(1-w)C_4 \\
 & + x(1-y)wc_7 + x(1-y)(1-w)c_8 \\
 & + (1-x)ywC_{11} + (1-x)y(1-w)C_{12} \\
 & + (1-x)(1-y)wC_{15} + (1-x)(1-y)(1-w)C_{16} \\
 U_S = & zU_{S_1} + (1-z)U_{S_2}
 \end{aligned} \tag{7}$$

$$F(z) = z(U_{S_1} - U_S) = z(1-z)(U_{S_1} - U_{S_2}) = z(1-z)C(x, y, w) \tag{8}$$

$$\begin{aligned}
 C(x, y, w) = & r_u - C_0 + \frac{C_l W_2}{\eta(\alpha-1)} \\
 & + xC_0\lambda_2 \\
 & + yW_2(1-\eta)\frac{C_l}{\eta(1-\alpha)} \\
 & + w(W_1 - W_2)\frac{C_l}{\eta(1-\alpha)} \\
 & + yw(W_1 - W_2)(1-\eta)\frac{C_l}{\eta(1-\alpha)} \\
 F'(z) = & (1-2z)C(x, y, w)
 \end{aligned} \tag{9}$$

Therefore, the adoption of the active participation strategy by SCP is jointly determined by the strategy choices of the other three actors, expected return, the water utilisation coefficient, the reserved water volume coefficient, and the level of subsidies provided by LCG. According to the stability analysis of the replicator dynamics equation, the strategy probability reaches a stable state when the conditions $F(z) = 0$ and $F'(z) < 0$ are satisfied.

Proposition 3. When $x < x_3$, the stable strategy of SCP is the opportunistic participation strategy. When $x > x_3$, the stable strategy of SCP is the active participation strategy. When $x = x_3$, the stable strategy of SCP is indeterminate. The threshold x_3 is defined as $x_3 = \frac{-r_u + C_0 - \frac{C_l W_2}{\eta(\alpha-1)} - yW_2(1-\eta)\frac{C_l}{\eta(1-\alpha)} - w(W_1 - W_2)\frac{C_l}{\eta(1-\alpha)} - yw(W_1 - W_2)(1-\eta)\frac{C_l}{\eta(1-\alpha)}}{C_0\lambda_2}$.

Proposition 3 indicates that SCP's willingness to invest is partly shaped by the strategy adopted by LCG. When LCG adopts the direct intervention strategy, SCP faces stronger institutional constraints and administrative control, making its participation behaviour more likely to evolve towards the opportunistic participation strategy. When LCG implements the incentive-oriented strategy, it enhances SCP's trust and willingness to participate, thereby encouraging a shift towards the active participation strategy. The threshold in Proposition 3 reflects the minimum return condition required for SCP to shift from opportunistic to active participation. In practice, this suggests that SCP's willingness to engage in irrigation water-rights trading is highly sensitive to the stability of expected return and the credibility of policy support. Without an adequate and predictable return structure, sustained participation is difficult to maintain.

Based on Proposition 3, the phase diagram of SCP's strategic choice is presented in Appendix A, Fig. A3.

Corollary 3.1. SCP is more likely to adopt the active participation strategy when expected return are high, subsidies are substantial, and the water utilisation coefficient is relatively high. By contrast, a larger reserved water volume coefficient increases the likelihood that SCP shifts towards the opportunistic participation strategy.

Corollary 3.2. When the investment return of SCP, r_u , exceeds the threshold r_u^* , defined as $r_u^* = C_0 - xC_0\lambda_2 + \frac{C_i W_2}{\eta(1-\alpha)} - yW_2(1-\eta)\frac{C_i}{\eta(1-\alpha)} - w(W_2 - W_1)\frac{C_i}{\eta(1-\alpha)} + yw(W_2 - W_1)(1-\eta)\frac{C_i}{\eta(1-\alpha)}$, SCP is more likely to adopt the active participation strategy. If the reserved water volume coefficient is relatively large, a higher expected return is required to sustain this strategic preference. Conversely, higher subsidies or an increase in the water utilisation coefficient reduce the threshold r_u^* , thereby increasing the likelihood that SCP will adopt the active participation strategy.

Analysis of evolutionarily stable strategies of IWU

The expected payoffs of IWU under the active water-saving strategy and the passive water-saving strategy are denoted by U_{W_1} and U_{W_2} , respectively, with the average expected payoff given by U_W . The replicator dynamics equation governing IWU's behavioural strategy, together with its first-order derivative, is presented in Eqs. (11) and (12).

$$\begin{aligned} U_{W_1} &= xyzd_1 + xy(1-z)d_3 \\ &\quad + x(1-y)zd_5 + x(1-y)(1-z)d_7 \\ &\quad + (1-x)yzd_9 + (1-x)y(1-z)d_{11} \\ &\quad + (1-x)(1-y)zd_{13} + (1-x)(1-y)(1-z)d_{15} \\ U_{W_2} &= xyzd_2 + xy(1-z)d_4 \\ &\quad + x(1-y)zd_6 + x(1-y)(1-z)d_8 \\ &\quad + (1-x)yzd_{10} + (1-x)y(1-z)d_{12} \\ &\quad + (1-x)(1-y)zd_{14} + (1-x)(1-y)(1-z)d_{16} \\ U_W &= wU_{W_1} + (1-w)U_{W_2} \end{aligned} \quad (10)$$

$$F_w = w(U_{W_1} - U_W) = w(1-w)(U_{W_1} - U_{W_2}) = w(1-w)D(x, y) \quad (11)$$

$$\begin{aligned} D(x, y) &= (n+1)P_0(W_0 - W_1) + P_1(W_2 - W_0) + u(W_1 - W_2) \\ &\quad + x\lambda_3 C_i \\ &\quad + 2yP_f \\ F'(w) &= (1-2w)D(x, y, z) \end{aligned} \quad (12)$$

Therefore, the adoption of the active water-saving strategy by IWU is jointly determined by the repurchase price multiplier for water rights, water-use benefits, the level of subsidies, and service fee. According to the stability conditions of the replicator dynamics equation, IWU's strategy probability reaches a stable state when the conditions $F(y) = 0$ and $F'(y) < 0$ are satisfied.

Proposition 4. When $x < x_4$ and $y < y_4$, the stable strategy of IWU is the passive water-saving strategy. When $x > x_4$ and $y > y_4$, the stable strategy of IWU is the active water-saving strategy. When $x = x_4$ and $y = y_4$, the stable strategy of IWU is indeterminate. The thresholds x_4 and y_4 are defined as $x_4 = \frac{u(W_2 - W_1) - (n+1)P_0(W_0 - W_1) - P_1(W_2 - W_0) - 2yP_f}{\lambda_3 C_i}$ and $y_4 = \frac{u(W_2 - W_1) - (n+1)P_0(W_0 - W_1) - P_1(W_2 - W_0) - x\lambda_3 C_i}{2P_f}$.

Proposition 4 indicates that IWU's adoption of the active water-saving strategy is shaped by the strategic choices of LCG and IMU. When LCG adopts the incentive-oriented strategy and IMU implements the high-efficiency management strategy, IWU benefits from the repurchase mechanism, water-saving incentives, and improved irrigation

services, thereby becoming more likely to adopt the active water-saving strategy. When LCG adopts the direct intervention strategy and IMU implements the inert management strategy, irrigation service quality deteriorates and expected return from water saving decline, increasing the likelihood that IWU maintains the passive water-saving strategy. The threshold in Proposition 4 indicates the minimum net-benefit condition under which IWU will adopt active water-saving behaviour. In practical terms, IWU is unlikely to shift towards sustained water conservation unless the combined gains from repurchase mechanisms, subsidies, and reduced water-use costs exceed the benefits derived from conventional water-use behaviour.

Based on Proposition 4, the phase diagram of IWU's strategic choice is presented in Appendix A, Fig. A4.

Corollary 4.1. IWU is more likely to adopt the active water-saving strategy when LCG sets a higher repurchase price multiplier for water rights, provides substantial subsidies, and IMU charges relatively high service fee. By contrast, a higher level of irrigation water-use benefits increases the likelihood that IWU shifts towards the passive water-saving strategy.

Corollary 4.2. When the repurchase price multiplier for water rights set by LCG, n , exceeds the threshold n^* , defined as $n^* = \frac{-2yP_f - P_1(W_2 - W_0) - u(W_1 - W_2) - x\lambda_3 C_i - P_0(W_0 - W_1)}{P_0(W_0 - W_1)}$, IWU is more likely to adopt the active water-saving strategy. When irrigation water-use benefits are relatively high, LCG needs to increase the repurchase price multiplier accordingly to sustain IWU's active water-saving strategy. Conversely, an increase in service fee or in the subsidy rate reduces the threshold n^* , thereby increasing the likelihood that IWU will adopt the active water-saving strategy.

The proofs of the Propositions and Corollaries are provided in Appendix A.

Evolutionary stability analysis of the system

In the replicator dynamic system involving LCG, IMU, SCP, and IWU, the equilibrium points are obtained by setting $F(x) = 0$, $F(y) = 0$, $F(z) = 0$, and $F(w) = 0$. Furthermore, by constructing the Jacobian matrix in Eq. (13) and applying Lyapunov's first theorem to examine the eigenvalues, the stability of each equilibrium point is analysed (Tu et al., 2025). It should be emphasised that this procedure identifies only local asymptotic stability. Therefore, the analytical results in this section should be interpreted as characterising the local stability properties of candidate equilibria under given parameter conditions, rather than as proving global convergence of the full dynamic system. Given that asymptotically stable solutions in multi-agent evolutionary games are typically associated with strict Nash equilibria, and that mixed-strategy equilibria may not satisfy the stability requirements of asymmetric dynamic games, this study focuses on the following 16 pure-strategy equilibrium points. The 16 pure-strategy equilibrium points considered in this study are given as follows: $E_1 = (0, 0, 0, 0)$, $E_2 = (1, 0, 0, 0)$, $E_3 = (0, 1, 0, 0)$, $E_4 = (0, 0, 1, 0)$, $E_5 = (0, 0, 0, 1)$, $E_6 = (1, 1, 0, 0)$, $E_7 = (1, 0, 1, 0)$, $E_8 = (1, 0, 0, 1)$, $E_9 = (0, 1, 0, 1)$, $E_{10} = (0, 1, 1, 0)$, $E_{11} = (0, 0, 1, 1)$, $E_{12} = (1, 1, 0, 1)$, $E_{13} = (1, 1, 1, 0)$, $E_{14} = (1, 0, 1, 1)$, $E_{15} = (0, 1, 1, 1)$, and $E_{16} = (1, 1, 1, 1)$.

$$J = \begin{bmatrix} \frac{\partial F(x)}{\partial x} & \frac{\partial F(x)}{\partial y} & \frac{\partial F(x)}{\partial z} & \frac{\partial F(x)}{\partial w} \\ \frac{\partial F(y)}{\partial x} & \frac{\partial F(y)}{\partial y} & \frac{\partial F(y)}{\partial z} & \frac{\partial F(y)}{\partial w} \\ \frac{\partial F(z)}{\partial x} & \frac{\partial F(z)}{\partial y} & \frac{\partial F(z)}{\partial z} & \frac{\partial F(z)}{\partial w} \\ \frac{\partial F(w)}{\partial x} & \frac{\partial F(w)}{\partial y} & \frac{\partial F(w)}{\partial z} & \frac{\partial F(w)}{\partial w} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \quad (13)$$

We further identify four fundamental conditions for the local asymptotic stability of the system. These conditions are not merely mathematical requirements for equilibrium stability but also reflect the institutional foundations of market-based irrigation water-rights allocation, thereby providing an economic and institutional rationale for why certain strategic configurations are more likely to persist than others (Hofbauer and Sandholm, 2009). First, $C_f - f - t_1 - t_2 < 0$ indicates that, under the direct intervention strategy, the maintenance cost borne by LCG is lower than the combined fiscal returns generated by the reward-penalty mechanism and water resources taxes. In practical terms, this condition implies that government intervention can be sustained only when it remains fiscally affordable. Second,

Table 2 Possible equilibrium points and their stability conditions.

Equilibrium Point	Sign	Stability	Stability conditions
(0, 0, 0, 0)	(−, −, *, *)	ESS	$r_u - C_0 - \frac{C_i W_2}{\eta(1-\alpha)} < 0;$ $P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - u(W_2 - W_1) < 0;$
(0, 0, 1, 0)	(*, *, *, *)	ESS	$C_f - f - t_1 - t_2 + (L-1)C_0\lambda_2 < 0;$ $P_f + V_f + f - C_a - \theta \frac{W_2}{1-\alpha} < 0;$ $C_0 - r_u + \frac{C_i W_2}{\eta(1-\alpha)} < 0;$ $P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - u(W_2 - W_1) < 0.$
(0, 0, 0, 1)	(*, −, *, *)	ESS	$C_f - f - t_1 - t_2 + (L-1)C_i\lambda_3 < 0;$ $r_u - C_0 - \frac{C_i W_1}{\eta(1-\alpha)} < 0;$ $-P_0(1+n)(W_0 - W_1) - P_1(W_2 - W_0) + u(W_2 - W_1) < 0.$
(1, 0, 1, 0)	(*, *, *, *)	ESS	$C_f - f - t_1 - t_2 + (L-1)(C_0\lambda_2 + C_a\lambda_1) < 0;$ $C_a - P_f - V_f - f + \theta \frac{W_2}{1-\alpha} < 0;$ $-r_u + C_0 + \frac{C_i W_2}{1-\alpha} < 0;$ $2P_f + P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - u(W_2 - W_1) < 0.$
(1, 0, 0, 1)	(*, *, *, *)	ESS	$C_f - f - t_1 - t_2 + (L-1)(C_0\lambda_2 + C_i\lambda_3) < 0;$ $P_f + V_f + f - C_a - \theta \frac{W_1}{1-\alpha} < 0;$ $-r_u + C_0 + \frac{C_i W_1}{\eta(1-\alpha)} < 0;$ $-P_0(1+n)(W_0 - W_1) - P_1(W_2 - W_0) + u(W_2 - W_1) < 0.$
(0, 1, 1, 0)	(*, *, *, *)	ESS	$f - C_f + t_1 + t_2 - (L-1)C_0\lambda_2 < 0;$ $P_f + V_f + f - C_a(1-\lambda_1) - \theta \frac{W_2}{1-\alpha} < 0;$ $-r_u + C_0(1-\lambda_2) + \frac{C_i W_2}{\eta(1-\alpha)} < 0;$ $P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - u(W_2 - W_1) + C_i\lambda_3 < 0.$
(0, 0, 1, 1)	(*, *, *, *)	ESS	$f - C_f + t_1 + t_2 - (L-1)C_i\lambda_3 < 0;$ $P_f + V_e + f - C_a(1-\lambda_1) - \theta \frac{W_1}{1-\alpha} < 0;$ $r_u - C_0(1-\lambda_2) - \frac{C_i W_1}{\eta(1-\alpha)} < 0;$ $-P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - C_i\lambda_3 + u(W_2 - W_1) < 0.$
(1, 1, 0, 1)	(*, *, *, −)	ESS	$C_f - f - t_1 - t_2 + (L-1)(C_a\lambda_1 + C_0\lambda_2 + C_i\lambda_3) < 0;$ $C_a - P_f - V_f - f + \theta \frac{W_1}{1-\alpha} < 0;$ $-r_u + C_0 + \frac{C_i W_1}{1-\alpha} < 0;$ $-P_0(1+n)(W_0 - W_1) - 2P_f - P_1(W_2 - W_0) + u(W_2 - W_1) < 0.$
(1, 0, 1, 1)	(*, *, −, *)	ESS	$f - C_f + t_1 + t_2 - (L-1)(C_0\lambda_2 + C_i\lambda_3) < 0;$ $P_f + V_f + f - C_a(1-\lambda_1) - \theta \frac{W_1}{1-\alpha} < 0;$ $-P_0(1+n)(W_0 - W_1) - P_1(W_2 - W_0) + u(W_2 - W_1) - C_i\lambda_3 < 0.$
(0, 1, 1, 1)	(*, *, *, *)	ESS	$f - C_f + t_1 + t_2 - (L-1)(C_a\lambda_1 + C_i\lambda_3) < 0;$ $C_a(1-\lambda_1) - P_f - V_e - f + \theta \frac{W_1}{1-\alpha} < 0;$ $r_u - C_0(1-\lambda_2) - \frac{C_i W_1}{1-\alpha} < 0.$
(1, 1, 1, 1)	(*, *, *, −)	ESS	$f - C_f + t_1 + t_2 - (L-1)(C_0\lambda_2 + C_a\lambda_1 + C_i\lambda_3) < 0;$ $C_a(1-\lambda_1) - P_f - V_f - f + \theta \frac{W_1}{1-\alpha} < 0;$ $-r_u + C_0(1-\lambda_2) + \frac{C_i W_1}{1-\alpha} < 0.$

“+” denotes a positive value, “−” denotes a negative value, and “*” denotes an indeterminate sign.

$C_a - P_f - V_e - f + \theta \frac{W_1}{1-\alpha} > 0$ implies that, under the high-efficiency management strategy, IMU can offset its operating costs through service fee, external revenues, and government incentives. This condition suggests that efficient management is sustainable only when sufficient economic compensation is available to support continued performance improvement. Third, $-r_u + C_0(1-\lambda_2) + \frac{C_i W_1}{\eta(1-\alpha)} < 0$ indicates that SCP's expected return exceeds its effective participation cost after subsidies and operational conditions are taken into account. This condition reflects the minimum profitability requirement for sustained SCP participation and implies that long-term participation depends on a sufficiently attractive and predictable return structure. Fourth, $2P_f + C_i\lambda_3 + P_0(1+n)(W_0 - W_1) + P_1(W_2 - W_0) - u(W_2 - W_1) > 0$ means that IWU's net return from active water-saving behaviour exceeds that from passive water use. This condition represents the economic incentive basis for IWU's transition towards water conservation. In practical terms, this suggests that water-saving and trading mechanisms can be sustained only when service-fee arrangements, repurchase incentives, and subsidies jointly compensate for the opportunity cost of reduced water use. Taken together, these conditions characterise the institutional and economic requirements under which specific equilibria may remain locally stable.

Given the above parameter settings and stability conditions, each pure-strategy equilibrium point is substituted into the Jacobian matrix of the replicator dynamic system to obtain the corresponding eigenvalues (Appendix A, Table A1). According to Lyapunov's first theorem, if all eigenvalues of the Jacobian matrix associated with a given equilibrium point have negative real parts, that equilibrium point is locally asymptotically stable, and the corresponding strategy profile can be regarded as an evolutionarily stable strategy (ESS) (Liu and Li, 2023). The results indicate that, among the 16 pure-strategy equilibrium points, 11 possible equilibrium points satisfy the stability conditions (Table 2).

Based on the set of locally stable pure-strategy equilibria identified above, the market-based allocation process may be interpreted through four stylised evolutionary states, namely the initial, development, formation, and maturation phases. This classification provides an institutional interpretation of the locally stable strategy profiles rather than describing a unique global transition path applicable to all initial conditions and parameter regions.

Initial phase. At this stage, all participating actors exhibit cautious or passive attitudes towards water-rights trading, and no effective coordination mechanism has yet emerged. LCG predominantly adopts the direct intervention strategy to maintain basic operational order, and the system evolves towards the equilibrium point $(0, 0, 0, 0)$, reflecting an initial state characterised by a low overall level of participation.

Development phase. As cognitive awareness improves and institutional arrangements develop, SCP and IWU begin to show stronger participation incentives, and the system evolves towards single-actor stable equilibrium points, including $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$. As the system moves into the middle and later stages of development, diversified bilateral partnerships begin to emerge. SCP and IMU coordinate to form cooperative configurations such as $(0, 1, 1, 0)$ and $(0, 0, 1, 1)$. In parallel, LCG shifts towards the incentive-oriented strategy, giving rise to configurations such as $(1, 0, 1, 0)$ and $(1, 0, 0, 1)$.

Formation phase. As multi-actor coordination mechanisms are gradually established, LCG's policy orientation steers the system towards distinct stable configurations. Under the incentive-oriented strategy, the system evolves towards $(1, 1, 0, 1)$ and $(1, 0, 1, 1)$, reflecting positive interactions among high-efficiency management by IMU, active participation by SCP, and active water-saving behaviour by IWU. By contrast, under the direct intervention strategy, the system evolves towards $(0, 1, 1, 1)$, representing a stable configuration dominated by administrative governance.

Maturation phase. LCG consistently adopts the incentive-oriented strategy, under which the water-rights trading mechanism becomes increasingly institutionalised, accompanied by improved IMU management efficiency, active SCP participation, and more active water-saving behaviour among IWU. Under parameter settings favourable to coordinated participation and within the simulated scenarios considered in this study, the system trajectory tends to converge towards the strategy profile $(1, 1, 1, 1)$, which can be interpreted as a desirable mature state of the market-based allocation mechanism.

Simulation and results analysis

The simulation analysis is grounded in the practical context of the Modernised Ecological Irrigation District Project in Litong District, Wuzhong City, Ningxia. As an important pilot initiative in modern ecological irrigation-district construction and water-rights reform, the project reflects key institutional features of agricultural water-saving reform, quota-based water management, and market-based water-rights trading in the water-scarce regions of northern China. The project involves the coordinated participation of LCG, IMU, SCP, and IWU, which corresponds closely to the four-party structure developed in this study and provides a realistic basis for examining multi-actor strategic interactions. The project also reveals several practical issues, including local fiscal pressure, the need to improve irrigation management efficiency, insufficient risk-sharing and return-guarantee mechanisms for SCP, and heterogeneous incentives for IWU water-saving behaviour. These issues are not unique to this case but are widely encountered in market-based irrigation water-rights reform across China. Taken together, these features make the project an appropriate case context for analysing how market-based allocation mechanisms operate under resource constraints and institutional transition.

Building on the theoretical analysis, we conduct numerical simulations of the proposed four-party evolutionary game model in MATLAB R2023b. The parameter values are calibrated with reference to project documents, official policy materials, field investigations, and expert judgement. Specifically, the values of W_0 , C_f , θ , C_t , P_f , n , V_f , C_a , f , t_1 , r_u , λ_1 , λ_2 and λ_3 are calibrated primarily with reference to the Implementation Plan for the PPP Project on the Construction of a Modern Ecological Irrigation District in Litong District, Wuzhong City and the corresponding investment plan. The values of P_0 , P_1 , C_b , C_0 , r_s , t_2 and η are derived primarily from official documents issued by the Ningxia Hui Autonomous Region and relevant local authorities. In addition, the value of u is informed by field surveys conducted in representative irrigation districts within Litong District and by interviews with local water authorities and personnel involved in the project. For parameters that cannot be directly observed or precisely quantified, such as L , benchmark values are assigned on the basis of expert consultation. The baseline parameter values are set as follows: $P_0 = 0.071$, $P_1 = 0.213$, $W_0 = 3.91$, $W_1 = 3$, $W_2 = 5$, $n = 15.21$, $C_a = 0.332$, $C_b = 0.8398$, $C_0 = 1.9596$, $C_t = 0.0305$, $P_f = 0.2567$, $C_i = 0.02$, $C_f = 0.04$, $f = 0.03866$, $t_1 = 0.143106$, $t_2 = 0.2346$, $r_u = 3.476$, $r_s = 0.01$, $V_f = 0.3476$, $V_e = 0$, $L = 2$, $\lambda_1 = 0.1$, $\lambda_2 = 0.4$, $\lambda_3 = 0.798$, $\theta = 0.0305$, $\alpha = 0.2$, $u = 0.05$, and $\eta = 0.586$. Among these parameters, the return parameter V_e for SCP under the opportunistic participation strategy is set to 0, reflecting the assumption that no direct return is generated when SCP does not engage in investment or management activities. The symbols, values, data sources, construction methods, and confidence levels of all parameters are reported in Appendix A, Table A2. Given the relatively low calibration confidence associated with L , additional stress tests are conducted under alternative values of L to examine the robustness of the main results, and the corresponding results are also reported in Appendix A, Fig. A5. The selected baseline parameter set satisfies the basic stability conditions of the identified evolutionarily stable strategies, thereby ensuring consistency between the numerical simulations and the theoretical framework.

Based on the baseline parameter setting, we numerically examine the dynamic properties of the four-party evolutionary game and assess the local stability of the pure-strategy equilibria by analysing their associated eigenvalues. The results show that $(1, 1, 1, 1)$ is locally asymptotically stable under the calibrated baseline parameter setting adopted in this study. In contrast, the local stability of the other candidate equilibria depends on more restrictive parameter combinations. Thus, within the calibrated parameter region considered here, $(1, 1, 1, 1)$ can be regarded as the dominant locally stable outcome in the baseline scenario.

Initial evolutionary path of the game system

Based on the four-party evolutionary game model, we simulate the initial evolutionary trajectories and visualise the three-dimensional phase paths in the (x, y, z) and (y, z, w) spaces. As shown in Appendix A, Fig. A6, although the initial strategy probability distributions of the agents are relatively dispersed, under the baseline parameter setting, the system trajectories progressively shift from low-probability regions towards high-probability regions and tend to move towards $(x, y, z, w) = (1, 1, 1, 1)$. The numerical simulation results further support the internal consistency and plausibility of the model specification. When LCG adopts the incentive-oriented strategy, IMU implements the high-efficiency management strategy, SCP chooses the active participation strategy, and IWU adopts the active water-saving strategy, the resulting payoff structure becomes more conducive to strategic coordination and the emergence of a locally stable outcome. This result indicates that the market-based allocation of irrigation water rights does not arise automatically from trading activities alone but depends on the coordinated interaction of government incentives, efficient management, SCP participation, and users' water-saving responses.

Effects of initial strategy probabilities on the evolutionary outcomes of the game system

To analyse the effects of initial strategy probabilities on system stability, we first simulate how variations in the initial strategy probability of a focal actor affect its evolutionary trajectory. In each simulation, the initial strategy probability of one focal actor is set to 0.1, 0.3, 0.5, 0.7, or 0.9, while the initial probabilities of the remaining three actors are fixed at 0.5. The simulation results are presented in Appendix A, Fig. A7, where the x-axis represents

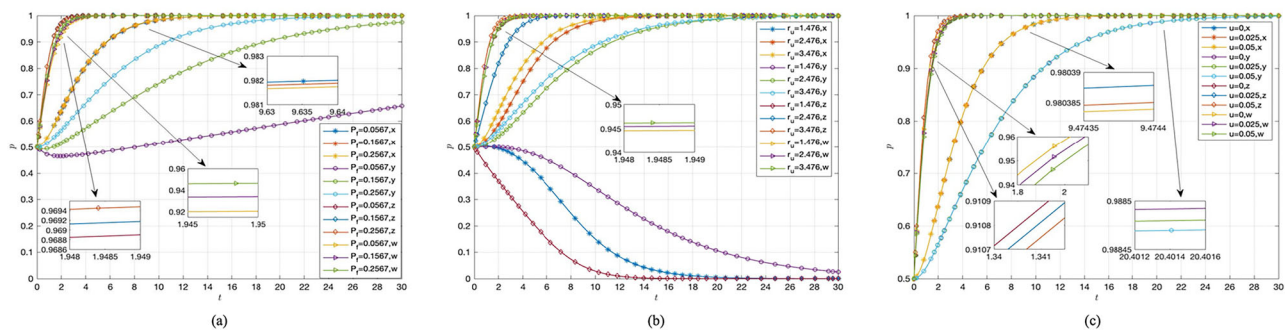


Fig. 4 Effects of fundamental factors on evolutionary game dynamics: **a** service fee P_f ; **b** expected return r_u ; **c** irrigation water-use benefits u .

evolutionary time t , and the y-axis denotes the strategy probability of the focal actor during the evolutionary process. The results indicate that, under the given parameter conditions, the strategy probabilities of all four actors eventually converge to stable states, although their convergence rates vary with the initial probabilities. LCG and IMU are the most sensitive to initial conditions, as lower initial probabilities lead to slower increases in their strategy probabilities and require more time to approach equilibrium. SCP and IWU are less sensitive to initial conditions, with their strategies converging more rapidly even when their initial probabilities are low. This result suggests that, within the simulated scenarios considered in this study, the early stage of reform exhibits path dependence in adjustment speed. Variations in initial conditions mainly affect the convergence speed and evolutionary trajectories, while leaving the direction of convergence towards the same locally stable outcome unchanged under the baseline scenario. Therefore, insufficient initial commitment from LCG or IMU may delay the formation of an effective market-based allocation mechanism.

Building on the preceding analysis, we further perform numerical simulations to examine how variations in the initial strategy probability of each actor affect the evolutionary trajectories of the other three actors. The results indicate that variations in actors' initial strategy probabilities mainly affect the speed of convergence towards a stable state, without altering the system's eventual evolutionary direction or equilibrium outcome. The corresponding simulation procedures and numerical results are reported in Appendix A.

Single-factor sensitivity analysis of the evolutionary game system

To systematically characterise how variations in key parameters affect the evolutionary trajectories and strategic stability of the water-rights trading system, we conduct a single-factor sensitivity analysis under the baseline parameter setting and equilibrium conditions. Specifically, the impacts of key parameter variations are examined across three dimensions, namely fundamental factors, policy-related factors, and management-technical factors. The fundamental factors include the service fee P_f , expected return r_u , and irrigation water-use benefits u . The policy-related factors include reward-penalty intensity f , the water-rights repurchase price multiplier n , water resources taxes t_1 and t_2 , fiscal subsidies λ_1 , λ_2 , and λ_3 , and reputational benefits L . The management-technical factors include the water utilisation coefficient η and the reserved water volume coefficient α . Some directional effects are already implied by the sign structure of the payoff functions. The following analysis therefore moves beyond these monotonic effects and focuses on the heterogeneous convergence patterns, cross-actor transmission effects, and dynamic responses generated by the four-party evolutionary system.

Effects of fundamental factors

Service fee represents cost compensation for IMU in providing water-dispatching, conveyance, distribution, and maintenance services. Because the positive effect of P_f on IMU has already been incorporated into the payoff structure, the analysis focuses on the heterogeneous convergence patterns and cross-actor transmission effects

associated with changes in P_f . Accordingly, three levels of P_f , namely $\{0.0567, 0.1567, 0.2567\}$, are specified for the numerical simulations. As shown in Fig. 4(a), increases in P_f lead to observable differences in the evolutionary trajectories and convergence speeds of the actors within the system. Overall, the convergence speeds towards stable strategies can be ranked, from fastest to slowest, as SCP, IWU, LCG, and IMU. Among the four actors, IMU exhibits the highest sensitivity to variations in P_f . When P_f is relatively low, heightened cost-recovery pressure causes IMU's evolutionary trajectory to exhibit an early-stage inflection point, indicating a phase of strategic delay. As the evolutionary process unfolds, IMU's behaviour gradually shifts from a low-sensitivity regime to a high-sensitivity regime and ultimately converges to a stable strategy. At higher levels of P_f , the magnitude of the inflection point diminishes, accompanied by faster convergence. By contrast, SCP and IWU respond more smoothly to variations in P_f , and their convergence speeds exhibit a weak positive association with P_f . Conversely, LCG exhibits a relative slowdown in convergence speed as P_f increases, suggesting a weak negative association between policy incentive effectiveness and P_f . In practical terms, P_f influences system evolution primarily through the IMU management channel. Lower P_f may weaken IMU's cost-recovery capacity and delay its transition towards high-efficiency management, whereas moderate increases can strengthen its operational incentives. However, adjustments to P_f should be coordinated with complementary policy incentives to avoid weakening the responsiveness of incentive-oriented governance.

The expected return of SCP constitutes a direct economic incentive for active participation. Because the positive effect of r_u on SCP participation has already been incorporated into the payoff structure, the analysis focuses on whether variations in r_u generate heterogeneous convergence patterns and spillover effects within the coupled system. Accordingly, three representative levels of r_u , namely $\{1.476, 2.476, 3.476\}$, are specified for the numerical simulations. As illustrated in Fig. 4(b), increases in r_u lead to heterogeneous evolutionary trajectories and convergence speeds across actors. Overall, SCP and IWU exhibit relatively rapid convergence, LCG shows a moderate convergence rate, and IMU converges most slowly. Further analysis indicates that when r_u remains at a low level, the evolutionary probabilities of SCP, LCG, and IMU decline rapidly and converge towards zero, reflecting a low-return decay pattern. As r_u increases, the strategic dynamics of these three actors reverse from decline to growth and eventually stabilise at a high-probability equilibrium. By contrast, IWU exhibits relatively weak sensitivity to changes in r_u , with a comparatively smooth strategic trajectory. Notably, once r_u exceeds a certain threshold, the convergence speed of SCP shifts from being lower than that of IWU to exceeding it. This result highlights the central role of r_u in sustaining SCP participation and stabilising the market-based evolution. When r_u is insufficient, SCP has limited incentives to participate actively, thereby constraining the strategic adjustments of LCG and IMU. Once r_u reaches a sufficient level, SCP participation becomes more stable and generates positive spillovers to the other actors, helping the system move beyond a low-participation trajectory. These findings suggest that the stability of r_u is essential for strengthening SCP participation incentives and avoiding a low-participation lock-in of the market mechanism.

Irrigation water-use benefits refer to the return obtained by IWU from conventional irrigation. In the payoff structure, a higher u directly weakens IWU's incentive to engage in water-saving behaviour and water-rights trading. Therefore, this section focuses on the dynamic responses of different actors to variations in u , rather than on the monotonic effect already implied by the payoff structure. Three levels of u , namely 0, 0.025, 0.05, are specified for numerical simulations. As illustrated in Fig. 4(c), the convergence speeds of actors towards a stable state can be ranked, from fastest to slowest, as SCP, IWU, LCG, and IMU. As u increases, the convergence speeds of LCG, IMU, SCP, and IWU all exhibit a slight downward trend. The weak negative effect of u suggests that higher returns from conventional irrigation may slow actors' strategic adjustment. When u remains high, IWU has weaker incentives to alter existing water-use behaviour, while the other actors face less pressure to optimise the trading mechanism. In practice, stronger external incentives may be needed to offset the inertia created by established production-return structures.

Effects of policy-related factors

The reward-penalty mechanism implemented by LCG is a key institutional instrument for guiding IMU's policy implementation and improving water-use management. Since the effects of f on LCG and IMU have been

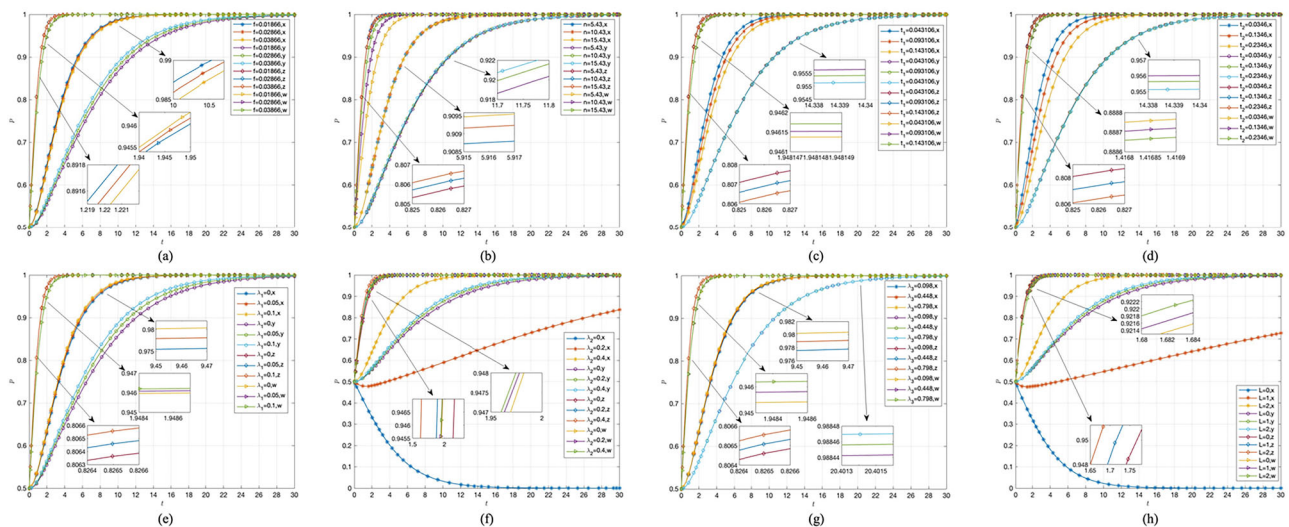


Fig. 5 Effects of policy-related factors on evolutionary game dynamics: **a** reward-penalty intensity f ; **b** repurchase price multiplier n ; **c** water resources tax t_1 ; **d** water resources tax t_2 ; **e** subsidy rate λ_1 ; **f** subsidy rate λ_2 ; **g** subsidy rate λ_3 ; **h** reputational benefits L .

incorporated into the payoff structure, the following analysis focuses on heterogeneous convergence responses and cross-actor transmission effects caused by variations in f . Accordingly, three levels of f , namely $\{0.01866, 0.02866, 0.03866\}$, are specified for numerical simulations. As shown in Fig. 5(a), increases in f generate heterogeneous evolutionary trajectories and convergence speeds among the actors. Overall, SCP and IWU converge most rapidly towards stable states, LCG exhibits a moderate convergence rate, and IMU converges most slowly, indicating actor-specific sensitivity to f . Further analysis indicates that IMU is most sensitive to variations in f . As the level of f imposed by LCG on IMU increases, the probability of IMU adopting the high-efficiency management strategy rises, accompanied by faster convergence. As regulation of IMU intensifies, LCG makes more conservative strategic adjustments, reflected in a slight decline in its probability of adopting the incentive-oriented strategy and slower convergence. For IWU, increases in f indirectly promote water-saving behaviour, indicating a weak positive association. For SCP, however, a higher f increases investment uncertainty, leading to more cautious investment decisions and a weak negative association. From a practical perspective, the reward-penalty mechanism shapes system evolution mainly through the IMU management channel. Increasing f can facilitate IMU's transition towards high-efficiency management and indirectly encourage IWU's water-saving behaviour, while excessive reliance on this instrument may delay LCG's strategic adjustment and weaken SCP's incentive to participate. Therefore, f should be maintained within a moderate range and coordinated with complementary incentives to strengthen managerial discipline without discouraging market participation.

The repurchase-price mechanism provides a direct economic incentive for IWU to save water and participate in water-rights trading. Because the effect of n on IWU's water-saving returns has been incorporated into the payoff structure, the following analysis focuses on the dynamic responses of different actors to variations in n . Accordingly, three levels of n , namely $\{5.21, 10.21, 15.21\}$, are specified for the numerical simulations. As shown in Fig. 5(b), the convergence speeds of actors' strategies towards stable states rank, from fastest to slowest, as SCP, IWU, LCG, and IMU. Among the actors, IWU is most sensitive to variations in n . As n increases, higher trading returns from water saving increase the probability of IWU adopting an active water-saving strategy and accelerate its convergence. By contrast, LCG, IMU, and SCP show relatively limited responses, with only marginal increases in strategy-selection probabilities and speeds. This result indicates that the repurchase-price mechanism operates mainly as a targeted incentive for IWU. A higher n increases the expected return from water saving and accelerates IWU's transition towards active water-saving behaviour, while its effects on LCG, IMU, and SCP remain relatively limited.

Water resources taxes serve as regulatory instruments through which LCG shapes water allocation and water-use behaviour. In the payoff structure, t_1 and t_2 directly affect LCG's payoff by altering its tax-related returns and may

further influence other actors' strategic adjustments through the coupled dynamics of the system. To examine these effects, numerical simulations are conducted under three levels of t_1 , namely $\{0.043106, 0.093106, 0.143106\}$, and three levels of t_2 , namely $\{0.0346, 0.1346, 0.2346\}$. Figure 5(c) shows the effects of the water resources tax t_1 levied by LCG on IMU, while Fig. 5(d) shows the effects of the water resources tax t_2 levied on IWU. Overall, SCP and IWU exhibit relatively rapid convergence; LCG shows a moderate convergence rate, and IMU converges most slowly. Further analysis indicates that LCG is most sensitive to variations in both t_1 and t_2 , with higher tax rates slowing its convergence. IWU exhibits differentiated responses to the two tax instruments, with moderate increases in t_1 indirectly encouraging the adoption of an active water-saving strategy through pressure transmission, whereas increases in t_2 raise water-use costs and weaken incentives for water saving and trading, resulting in slightly slower convergence. IMU and SCP exhibit only marginal declines in convergence speed in response to changes in tax rates. From a practical perspective, water resources taxes exert differentiated effects across actors. A moderate tax imposed on IMU can transmit management pressure and indirectly promote IWU's water-saving behaviour, whereas a higher tax imposed on IWU may increase water-use costs and weaken its incentives for water saving and trading. Therefore, tax design should be actor-specific and carefully calibrated to balance regulatory pressure with behavioural incentives.

Fiscal subsidies provided by LCG are a key policy lever shaping the system's evolutionary dynamics. Since the direct incentive effects of λ_1 , λ_2 , and λ_3 on the corresponding recipient actors have been incorporated into the payoff structure, the following analysis focuses on heterogeneous convergence patterns and cross-actor transmission effects induced by subsidy adjustments. Accordingly, three levels of λ_1 , namely $\{0, 0.05, 0.1\}$, three levels of λ_2 , namely $\{0, 0.2, 0.4\}$, and three levels of λ_3 , namely $\{0.198, 0.498, 0.798\}$, are specified for the numerical simulations. Figure 5(e) shows the effects of λ_1 , Fig. 5(f) shows the effects of λ_2 , and Fig. 5(g) shows the effects of λ_3 . Overall, changes in subsidy rates affect convergence speed and stability heterogeneously across actors. When λ_1 and λ_3 vary, SCP and IWU converge relatively quickly, LCG converges at a moderate rate, and IMU converges most slowly. As λ_2 increases, LCG's strategic evolution exhibits a staged pattern, shifting from the direct intervention strategy at low subsidy levels to the incentive-oriented strategy at higher subsidy levels, with its convergence speed gradually surpassing that of IMU. Further analysis indicates that IMU is most sensitive to variations in λ_1 , while increases in λ_2 elicit the strongest response from LCG, followed by IMU. Increases in λ_3 exert relatively limited influence on the evolutionary processes of all actors, only modestly improving convergence speed and stability. These results suggest that fiscal subsidies have actor-specific effects. Subsidies provided to IMU mainly support high-efficiency management, while subsidies provided to SCP facilitate LCG's transition from direct intervention to incentive-oriented governance. Subsidies provided to IWU only modestly improve convergence, indicating that direct cost compensation alone produces limited dynamic effects. Therefore, fiscal subsidies should be targeted towards strengthening management capacity and stabilising SCP participation.

In water-rights trading, reputational benefits represent a non-economic incentive that may affect LCG's strategic adjustment and thereby shape the system's evolutionary dynamics. Because the direct effect of L on LCG's payoff has been incorporated into the payoff structure, the following analysis focuses on the dynamic responses and cross-actor transmission effects induced by variations in L . Accordingly, three levels of L , namely $\{1, 1.5, 2\}$, are specified for the numerical simulations, as shown in Fig. 5(h). The results indicate that variations in L have heterogeneous effects on the speed and stability of strategy evolution across actors. Overall, SCP and IWU exhibit relatively rapid convergence, while LCG's strategic evolution displays a staged pattern. When L is low, LCG tends to adopt the direct intervention strategy. As L increases, LCG's evolutionary trajectory gradually shifts towards the incentive-oriented strategy, and its convergence speed rises from initially lagging behind that of IMU to gradually surpassing it. Further analysis indicates that LCG is most sensitive to variations in L , followed by IMU, and that increases in L accelerate the convergence of both actors towards stable states. SCP and IWU respond more gradually to variations in L , exhibiting weak positive associations. This result suggests that L is an important non-economic driver of LCG's strategic adjustment. A higher L makes incentive-oriented governance more attractive to LCG and accelerates the strategic adjustment of both LCG and IMU. The weaker responses of SCP and IWU indicate that L operates mainly through the government–management channel rather than through direct incentives for market participants.

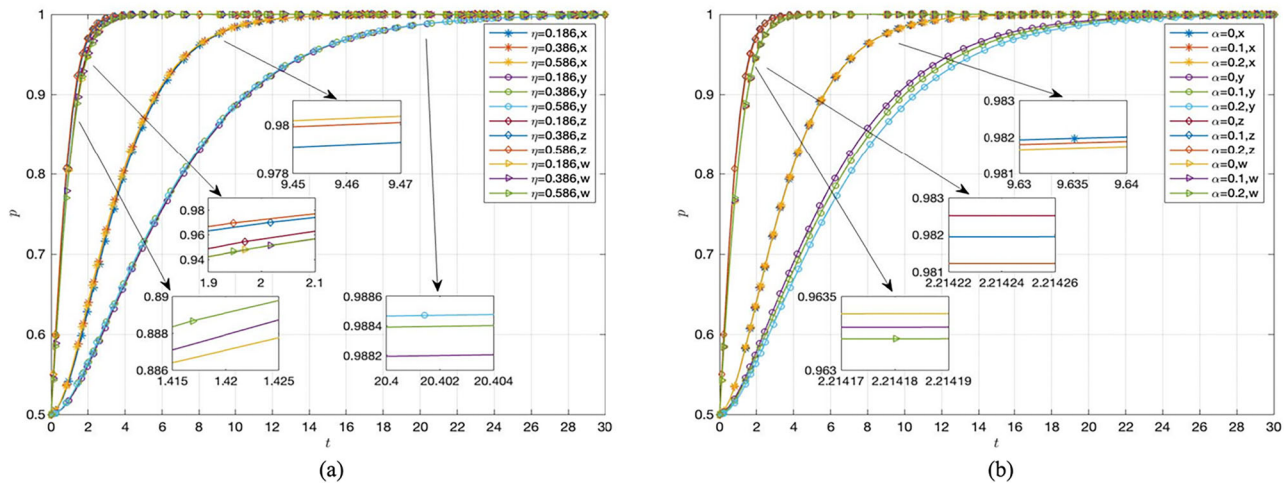


Fig. 6 Effects of management-technical factors on evolutionary game dynamics: **a** water utilisation coefficient η ; **b** reserved water volume α .

Therefore, information disclosure, performance evaluation, and public oversight should be strengthened to enhance the effect of L and support incentive-oriented governance.

Effects of management-technical factors

The water utilisation coefficient is a key indicator of IMU's management efficiency. To examine the effects of IMU's water utilisation coefficient η on IWU's participation in water-rights trading, three representative levels of η , namely $\{0.186, 0.386, 0.586\}$, are considered in the simulations. As shown in Fig. 6(a), the convergence speeds of actors' strategies towards stable states rank, from fastest to slowest, as SCP, IWU, LCG, and IMU. The simulations show that, as η increases, water-use efficiency improves and the convergence speeds of LCG, IMU, SCP, and IWU increase slightly. From a practical perspective, a higher η can moderately improve system responsiveness by reducing resource losses and strengthening IMU's allocation capacity. Although this factor alone is insufficient to drive the market-based transition, it provides technical support for accelerating the convergence of the market-based allocation mechanism.

The reserved water volume reflects IMU's water-supply security constraints and allocation flexibility. Because its direct effect is already reflected in the payoff structure, the following analysis focuses on the dynamic responses of different actors to variations in α . Accordingly, three levels of α , namely $\{0, 0.1, 0.2\}$, are specified for the numerical simulations. As shown in Fig. 6(b), the convergence speeds of actors' strategies towards stable states rank, from fastest to slowest, as SCP, IWU, LCG, and IMU. The results indicate that IMU is most sensitive to variations in α . As α increases, constraints on allocable water intensify and operational uncertainty rises, markedly slowing IMU's convergence towards the high-efficiency management strategy. The responses of LCG, SCP, and IWU to variations in α are relatively weak, with only marginal reductions in convergence speed, indicating a weak negative association. This result suggests that α mainly affects IMU by limiting flexible water allocation and increasing operational uncertainty, thereby slowing its convergence towards high-efficiency management. Its effects on LCG, SCP, and IWU remain relatively weak. Therefore, α should be maintained within a reasonable range to balance water-supply security, management flexibility and trading efficiency.

Two-factor sensitivity analysis of the evolutionary game system

Although single-factor sensitivity analysis helps identify the effects of individual parameters, system stability in a four-party evolutionary game cannot be fully explained by isolated parameter changes. Instead, it depends on the joint effects of related parameters and the coupled strategic adjustments of LCG, IMU, SCP, and IWU. Therefore, a two-

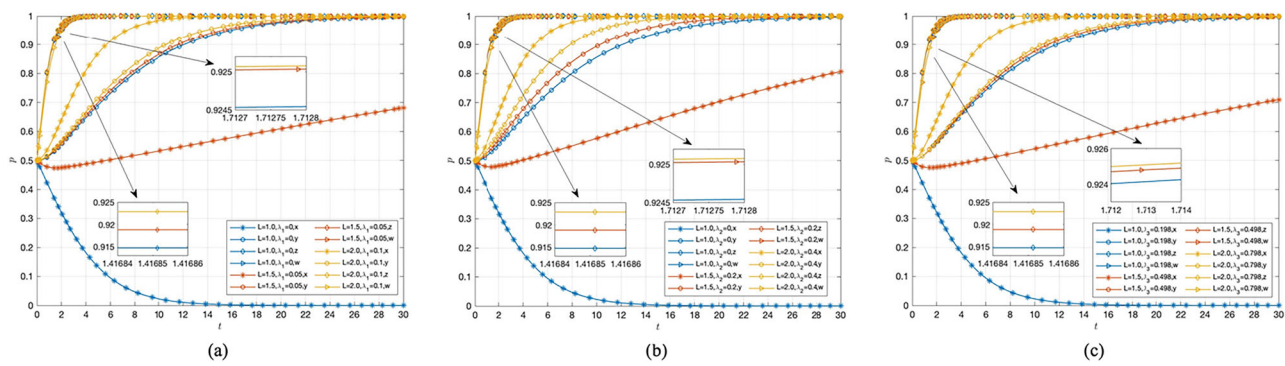


Fig. 7 Joint effects of LCG-related parameters on evolutionary game dynamics: **a** λ_1 and L ; **b** λ_2 and L ; **c** λ_3 and L .

factor sensitivity analysis is conducted using representative parameter combinations linked to the core decisions of the four actors. This analysis examines whether joint parameter changes generate reinforcing, offsetting, or asymmetric effects on evolutionary trajectories, convergence speed, and stable equilibria.

Joint effects of LCG-related parameters

Fiscal subsidies and reputational benefits capture, respectively, the economic incentives and social constraints faced by LCG, and their interaction shapes governance orientation and the intensity of policy incentives. To identify the interaction mechanism between these two categories and their combined effects on system evolution, a two-factor sensitivity analysis is conducted. In this analysis, the parameters are specified as $\lambda_1 \in \{0, 0.05, 0.1\}$, $\lambda_2 \in \{0, 0.2, 0.4\}$, $\lambda_3 \in \{0.198, 0.498, 0.798\}$ and $L \in \{1, 1.5, 2\}$. The analysis examines the convergence speed and evolutionary trajectory of each actor's strategy under the joint variation of these two categories of factors.

Panels (a)–(c) in Fig. 7 illustrate the joint effects of λ_1 and L , λ_2 and L , and λ_3 and L , respectively, on the actors' evolutionary dynamics. When both subsidy rates and L remain low, LCG tends to adopt the direct intervention strategy, while the other actors gradually converge to stable states. As both factors increase, LCG gradually shifts towards the incentive-oriented strategy, and its convergence speed rises from initially lagging behind that of IMU to eventually surpassing it. Under the joint effects of subsidy rates and L , IMU shows a stronger response and faster convergence. SCP and IWU respond more slowly, showing only weak positive associations. Overall, the joint effects of subsidy rates and L are directionally consistent, although their magnitudes differ across actors. IMU is most sensitive to the joint variation in λ_2 and L , followed by λ_1 and L , and is least responsive to the combination of λ_3 and L . Compared with the single-factor sensitivity analysis, the two-factor interactions generate more complex dynamics in system responsiveness and strategic trajectories. The introduction of L transforms fiscal subsidies from economic compensation instruments into composite policy tools that integrate performance incentives and reputational constraints. This transformation promotes LCG's strategic shift from direct intervention towards incentive-oriented governance and broadens policy incentives beyond economic compensation. The synergy between fiscal subsidies and L strengthens policy responsiveness and strategic adjustment efficiency, enabling the system to reach stable convergence more rapidly than under single-factor changes. From a practical perspective, fiscal subsidies alone are insufficient to sustain market-based irrigation water-rights trading. Their effectiveness depends on reputational incentives that reinforce LCG's policy continuity and governance credibility. The stronger response of IMU further indicates that irrigation management is the main channel through which government incentives affect system performance. Therefore, improving irrigation water-rights trading requires greater financial support, stronger performance assessment, information disclosure, and public oversight.

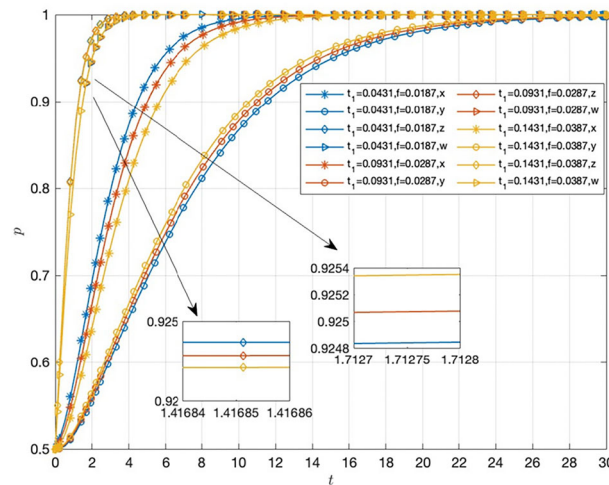


Fig. 8 Joint effects of IMU-related parameters on evolutionary game dynamics.

Joint effects of IMU-related parameters

The water resources tax and reward-penalty intensity imposed on IMU are parallel regulatory instruments employed by LCG, reflecting fiscal constraints and institutional incentives, respectively. Given that their interaction effects may be synergistic or offsetting, the parameters are specified as $t_1 = \{0.0431, 0.0931, 0.1431\}$ and $f = \{0.0187, 0.0287, 0.0387\}$. Their joint variation is examined to assess its effects on each actor's convergence speed and evolutionary trajectory.

As shown in Fig. 8, under the joint variation of t_1 and f , the strategic evolution of each actor exhibits interaction patterns that differ from those observed under single-factor variations. When both parameters increase, the convergence speeds of LCG and SCP are slightly lower than those observed in the single-factor scenarios, while IWU's convergence accelerates. When t_1 varies alone, IMU shows a slight decline in convergence speed because of increased costs, while variation in f produces an upward trend as performance incentives strengthen. When the two factors interact, the marginal incentive effects of f outweigh the restraining effect of the rising tax burden, resulting in an overall increase in IMU's convergence speed. This pattern reflects a policy synergy characterised by incentive dominance. From a practical perspective, this result suggests that water resources taxes and reward-penalty mechanisms should be coordinated. Higher taxation alone may increase IMU's operating burden and slow its strategic adjustment. When combined with a stronger f , however, taxation may promote faster convergence towards efficient management by offsetting its adverse cost effects. Thus, LCG should combine restrictive and incentive-based instruments rather than relying solely on fiscal pressure. The faster convergence of IWU further indicates that such policy coordination can enhance water-saving responses. Therefore, a balanced regulatory framework that integrates moderate taxation with performance-based rewards and penalties would better support the stable operation of irrigation water-rights trading.

Joint effects of SCP-related parameters

SCP's expected return determine its willingness to participate in the water-rights trading market and its investment intensity, whereas IMU's water-use efficiency and reserved water volume reflect its technical capacity and water-supply security margin. In practice, IMU's technical capacity directly affects SCP's investment risk and return expectations, while SCP's capital input can in turn feed back into IMU's management system. By strengthening water-dispatching capacity, canal maintenance, and water-saving technological upgrading, this feedback can improve water-use efficiency and optimise the allocation of reserved water volumes. To examine the interaction effects between SCP's expected return and IMU's water utilisation coefficient and reserved water volume within the water-

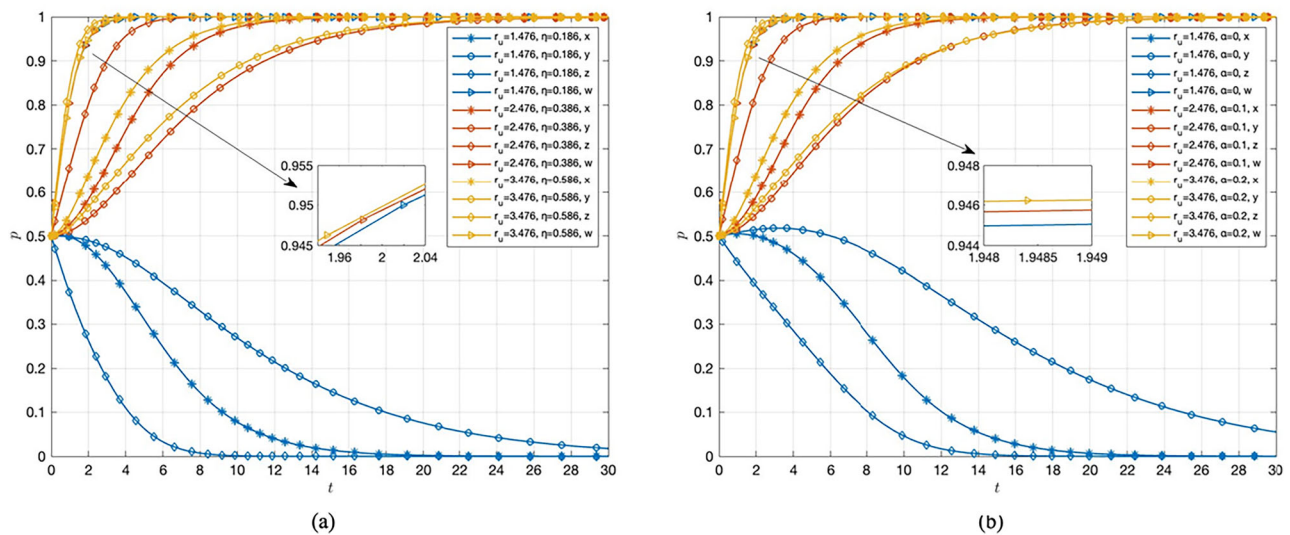


Fig. 9 Joint effects of SCP-related parameters on evolutionary game dynamics: **a** r_u and η ; **b** r_u and α .

rights trading system, a two-factor sensitivity analysis is conducted. Specifically, the parameters are specified as $r_u \in \{1.476, 2.476, 3.476\}$, $\eta \in \{0.186, 0.386, 0.586\}$, and $\alpha \in \{0, 0.1, 0.2\}$. Their joint variation is examined to assess differences in convergence speed and evolutionary trajectories across the actors.

Figure 9(a), (b) illustrate the joint effects of r_u and η , and of r_u and α , respectively, on the system's evolutionary dynamics. When r_u and either η or α increase simultaneously, IWU and SCP converge most rapidly towards stable states, followed by LCG, while IMU converges most slowly. When r_u and either η or α remain low, SCP, LCG, and IMU converge towards the opportunistic participation, direct intervention, and high-efficiency management strategies, respectively. By contrast, IWU continues to converge towards the active water-saving strategy. As r_u rises together with η or α , the strategic probabilities of SCP, LCG, and IMU shift from decline to growth and eventually stabilise at high levels. IWU responds more weakly to these joint changes, showing only a slight increase in convergence speed. Compared with the single-factor analysis, the two-factor interactions produce only modest changes in overall convergence, and the outcomes largely mirror those obtained when varying r_u alone. This suggests that r_u is the dominant economic driver of system evolution, whereas IMU's managerial and technical factors play mainly auxiliary and moderating roles. From a practical perspective, active SCP participation depends primarily on stable and attractive expected return. Improvements in IMU's water-use efficiency and reserved water allocation can reduce investment risks. Therefore, policy design should prioritise return stabilisation, with management and allocation improvements serving as supporting measures.

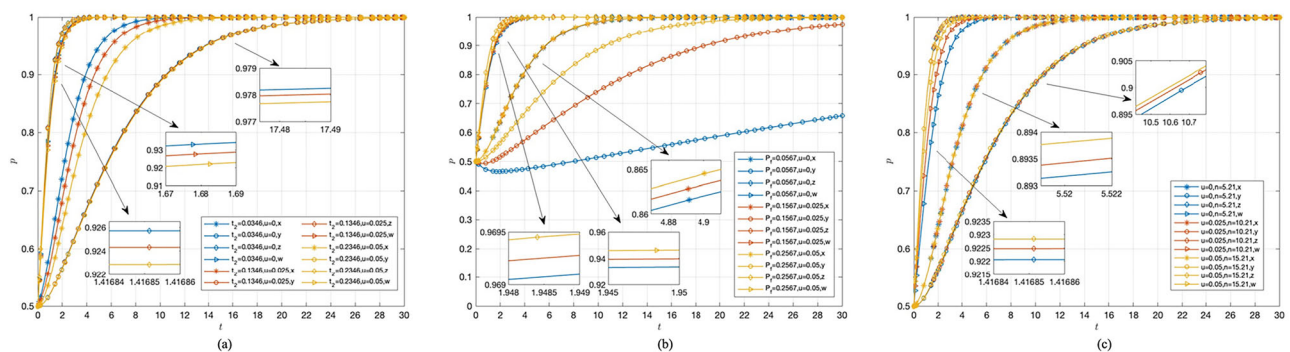


Fig. 10 Joint effects of IWU-related parameters on evolutionary game dynamics: **a** t_2 and u ; **b** P_f and u ; **c** u and n .

Joint effects of IWU-related parameters

The water resources tax imposed on IWU is an important economic lever for regulating agricultural water-use behaviour, while irrigation water-use benefits reflect the economic alignment between water resource inputs and agricultural outputs. Changes in the water resources tax burden directly alter the structure of irrigation returns, thereby affecting water-use benefits. Changes in water-use benefits, in turn, constrain the rationale of tax system design and influence its regulatory effectiveness. To examine the effects of their interaction on system evolution, a two-factor sensitivity analysis is conducted with $t_2 \in \{0.0346, 0.1346, 0.2346\}$ and $u \in \{0, 0.025, 0.05\}$. Their joint variation is examined to assess its effects on the convergence speed and evolutionary trajectory of each actor's strategy. As shown in Fig. 10(a), under the joint variation of t_2 and u , SCP and IWU converge most rapidly towards stable states, followed by LCG, while IMU converges most slowly. Further analysis indicates that when t_2 and u increase simultaneously, LCG shows the strongest strategic response, with its convergence speed towards a stable state declining markedly. The convergence speeds of IMU, SCP, and IWU decline only slightly. Compared with the single-factor sensitivity analysis, the two-factor interaction produces only limited changes in the system's overall convergence characteristics, with evolutionary trajectories resembling those observed when t_2 varies alone. This indicates that, in scenarios dominated by the fiscal constraint t_2 , the regulatory role of irrigation water-use benefits u remains relatively weak, and their joint variation has only limited influence on the system's dynamic stability and strategic evolution paths.

The service fee charged by IMU constitutes a cost constraint on IWU's water use, whereas irrigation water-use benefits reflect IWU's economic returns, with both jointly determining IWU's willingness to conserve water and to participate in water-rights trading. To examine the interaction mechanism between these two factors, a two-factor sensitivity analysis is conducted for the service fee P_f and the irrigation water-use benefit parameter u . Specifically, the parameters are specified as $P_f \in \{0.0567, 0.1567, 0.2567\}$ and $u \in \{0, 0.025, 0.05\}$. Their joint variation is examined to compare differences in convergence speed and evolutionary trajectories across actors' strategies. As shown in Fig. 10(b), when P_f and u increase simultaneously, SCP and IWU converge relatively rapidly towards stable states, followed by LCG, while IMU converges most slowly. Further analysis indicates that IMU's evolutionary trajectory exhibits stage-specific characteristics. When P_f and u remain low, IMU's strategic trajectory shows an early turning point and then gradually converges towards a stable state. As both parameters increase, IMU's convergence towards the high-efficiency management strategy accelerates. LCG, SCP, and IWU show relatively weak responses to changes in P_f and u , characterised by weak positive associations. Compared with the single-factor sensitivity analysis, the two-factor interaction introduces a degree of synergy in strategic evolution, shifting LCG's convergence trend from weakly negative to weakly positive. SCP's strategy convergence shows a slight positive association, as the weak positive effect of the service fee P_f marginally outweighs the negative constraining effect of the water-use benefit parameter u . These findings indicate that a reasonable P_f can strengthen IMU's cost-recovery capacity and promote high-efficiency management, while remaining compatible with IWU's affordability and supporting stable water-rights trading.

The repurchase price multiplier set by LCG represents an external incentive factor, while irrigation water-use benefits constitute an internal constraint. Changes in either factor alone are insufficient to fully capture the effects of internal-external interactions on the dynamic characteristics of the game system. Accordingly, a two-factor sensitivity analysis is conducted for u and n , specifying $u \in \{0, 0.025, 0.05\}$ and $n \in \{5.21, 10.21, 15.21\}$. Their joint variation is examined to assess differences in convergence speed and evolutionary trajectories across actors' strategies. As shown in Fig. 10(c), when u and n increase simultaneously, SCP and IWU converge most rapidly towards stable states, followed by LCG, while IMU converges most slowly. Further analysis indicates that IWU is most sensitive to the joint variation in these two factors, with its convergence speed increasing markedly. Compared with the single-factor sensitivity analysis, the two-factor interaction produces only limited changes in the convergence characteristics of each actor's strategy, with evolutionary trajectories resembling those observed when n varies alone. However, the convergence speeds of LCG, IMU, and SCP are slightly reduced, while IWU's convergence speed increases marginally. This result indicates that a higher u may reinforce conventional water use, while its combination with a higher

n can make water-saving gains more attractive and accelerate IWU's active participation. Therefore, LCG should adjust n according to users' actual u to compensate for the opportunity cost of water saving.

Discussion

Theoretical framework and interpretation of main findings

Compared with existing studies that mainly focus on static outcomes, pricing effects, or simplified actor relationships, the four-party evolutionary game framework developed in this study provides a more comprehensive explanation of the market-based allocation mechanism of irrigation water rights. By incorporating LCG, IMU, SCP, and IWU into a unified analytical framework, the model captures strategic interdependence among key stakeholders more fully than conventional three-actor frameworks and better reflects the institutional complexity of irrigation water-rights trading in China.

Methodologically, this study shifts the analytical focus from static comparisons of market performance to the dynamic process through which market-based allocation emerges, stabilises, or loses stability under bounded rationality. This approach explains which outcomes may occur and clarifies how, and under what conditions, different actors adjust their strategies and move towards specific equilibrium configurations. The results suggest that the market-based development of irrigation water-rights trading emerges from continuous strategic adjustment among multiple actors under changing incentive and constraint conditions, rather than from a single policy variable.

The main findings can be interpreted through the broader theoretical lens of incentive compatibility, institutional constraints, and multi-actor coordination in common-resource governance. The positive effects of fiscal subsidies and reputational benefits on LCG's incentive-oriented strategy suggest that LCG behaviour is shaped jointly by economic incentives and external accountability. IMU's sensitivity to service fee, reward-penalty intensity, and reserved water volume reflects the combined role of operational compensation and institutional discipline in management decision-making. SCP's strong dependence on expected return and management efficiency is consistent with the logic of risk–return matching in public–private cooperation. IWU's responsiveness to repurchase multipliers, service fee, and water resources taxes indicates that water-saving behaviour depends on whether conservation efforts can be translated into stable and observable economic returns. Overall, these findings suggest that the market-based allocation of irrigation water rights is governed by the coordinated interaction of incentives, constraints, and participation expectations across actors, rather than by any single policy instrument.

Bridging theory and practice

The four-party evolutionary game framework developed in this study provides a dynamic perspective for explaining how incentives, constraints, and participation expectations jointly shape the evolution of market-based irrigation water-rights allocation. Rather than treating water-rights trading as a static institutional arrangement, the framework highlights the behavioural mechanisms through which multiple actors, under bounded rationality, gradually move towards or away from market-based configurations.

Theoretically, the main findings are broadly consistent with common-resource governance and institutional incentive theory. The positive effects of fiscal subsidies, reputational benefits, expected return, and repurchase mechanisms indicate that market-based allocation is more likely to emerge when actors obtain clear and sustained benefits from participation, efficient management, and water-saving behaviour. The effects of reward-penalty intensity, taxation, and management conditions further suggest that market-based allocation requires institutional constraints, performance-oriented governance, and credible implementation mechanisms in addition to price signals. This implies that the effectiveness of irrigation water-rights trading depends on the coordinated alignment of economic incentives, managerial capacity, and behavioural responses across actors.

These theoretical implications are partly supported by practical observations from China's water-rights trading reforms. In existing pilot areas, clearer trading rules, compensation arrangements, and supporting incentives have been associated with stronger water-saving incentives and more active participation in water reallocation. Evidence from Ningxia and other northwestern irrigation districts further suggests that, under strict resource constraints, appropriately designed pricing and compensation mechanisms can reduce users' burdens while balancing agricultural, ecological, and other competing water demands (Yin et al. 2025; Qin et al. 2022). These observations are consistent with the model result that market-based allocation is more likely to evolve when pricing incentives, fiscal support, and compensation arrangements operate in a coordinated manner. International experience further supports the institutional logic identified in this study. Research on water markets in Australia, Spain, and other regions shows that clearly defined rights, transparent trading platforms, and effective information disclosure can improve water reallocation efficiency by strengthening property-rights incentives and reducing transaction uncertainty, while weak institutional enforcement, insufficient coordination, and high uncertainty tend to constrain market performance (Grafton et al. 2010; Palomo-Hierro et al. 2015; Calatrava and Martínez-Granados, 2018). This is consistent with the present model, in which stable market-based evolution depends on tradable rights and on the broader institutional environment that shapes trust, enforcement, and participation expectations.

Therefore, the value of the present framework lies in clarifying the underlying mechanisms through which multi-actor incentives and constraints jointly shape the evolution of market-based irrigation water-rights trading, rather than in reproducing all institutional and behavioural complexities observed in practice. In this way, the framework serves as an analytical bridge between institutional theory and practical reform design.

Distributional consequences, accountability risks, and ecological safeguards

Although coordinated incentives can promote market-based convergence, heterogeneity among IWU shapes their effects. Differences in farm size, cropping structure, irrigation dependence, water-saving capacity, access to information, and risk tolerance influence farmers' ability to respond to price signals and participate in water-rights trading (Zhang et al. 2019; Rong and Cheng 2024). Larger or more commercially oriented farmers are better positioned to convert water savings into tradable gains, while smallholders and farmers with inelastic irrigation demand may face higher adjustment costs and weaker bargaining power (Hadjigeorgalis, 2008; Danso et al. 2021). Consequently, repurchase prices, service fee, subsidies, and water resources taxes may generate uneven participation incentives and distributional effects. Without adequate metering, transparent information disclosure, and transaction-support mechanisms, water-rights trading may improve aggregate efficiency while marginalising relatively vulnerable users.

SCP participation may also create accountability risks. Although stable returns are necessary to sustain SCP participation, SCP may selectively serve profitable areas, transfer costs to water users, underinvest in long-term maintenance, or engage in opportunistic behaviour under weak monitoring (Iossa and Martimort, 2015). Contractual arrangements should clearly define investment obligations, maintenance responsibilities, service standards, disclosure requirements, risk-sharing rules, and exit mechanisms. Market-based allocation also requires robust ecological safeguards (Tisdell, 2001). Tradable water volumes should be determined only after ecological water demand, basic agricultural water security, groundwater protection, and regional water-resource carrying capacity have been secured. Thus, irrigation water-rights trading should be understood as a market mechanism embedded in institutional and ecological constraints, rather than as an unconstrained instrument for efficiency improvement.

Research limitations and future directions

Although this study provides a useful theoretical framework for understanding the market-based allocation of irrigation water rights, several limitations should be acknowledged. First, the model is developed under the assumption of bounded rationality and abstracts from information asymmetry, heterogeneous risk preferences, and cognitive biases. It is therefore better suited to explaining the general direction and underlying mechanisms of strategic evolution than to reproducing the full complexity of individual behavioural responses in practice. Second,

several policy-related and behavioural variables, such as subsidy rates, reputational benefits, expected return, and irrigation water-use benefits, are parameterised as deterministic factors. This treatment facilitates comparative analysis but cannot fully capture time-varying institutional adjustments, fiscal changes, technological progress, or external environmental disturbances. Third, the stability analysis focuses mainly on local asymptotic stability and therefore cannot fully capture path switching, transitions across multiple equilibria, or pronounced dynamic fluctuations under major external shocks.

These limitations point to several directions for future research. Future studies may incorporate spatial heterogeneity and regional differentiation into the evolutionary framework to examine how differences in resource endowments, management efficiency, and policy constraints shape market development across irrigation districts. Future studies could also introduce time-varying parameters and multi-stage dynamic settings to better capture policy feedback, institutional adjustment, and improvements in management over time. In addition, the integration of the theoretical framework with empirical evidence could be strengthened through broader case comparisons, multi-source datasets, more refined parameter estimation, and additional validation procedures, thereby improving the framework's explanatory power and external applicability.

Conclusions and policy implications

Conclusions

Improving the market-based allocation mechanism of water rights for irrigation water users in China is consistent with sustainable agricultural water management, helps alleviate imbalances between water supply and demand, mitigates ecological and environmental risks, and provides institutional support for high-quality agricultural development. This study develops a four-party evolutionary game model involving LCG, IMU, SCP, and IWU to explain the evolutionary dynamics and potential improvement pathways of market-based irrigation water-rights trading. The main findings are as follows.

- (1) The system admits up to 11 pure-strategy equilibrium points that are locally asymptotically stable under specific parameter conditions. Under the baseline parameter setting adopted in this study, the strategy combination (1, 1, 1, 1) emerges as the dominant locally stable outcome in the numerical simulations. Under this equilibrium, LCG primarily adopts an incentive-oriented strategy that enhances policy effectiveness. IMU maintains high-efficiency management and stable service provision. SCP remains actively participating in investment. IWU's strategy converges towards active water-saving behaviour.
- (2) The evolutionary trajectory and stability of the market-based irrigation water-rights trading system are jointly influenced by the initial strategy probabilities of LCG, IMU, SCP, and IWU. Differences in these initial probabilities mainly affect convergence speed and adjustment paths, thereby generating heterogeneous dynamic trajectories. However, within the baseline setting and the simulated range considered in this study, these differences do not alter the system's observed tendency to move towards the same locally stable outcome.
- (3) Single-factor sensitivity analysis indicates that parameter variations exert heterogeneous effects on the system's evolutionary dynamics. Increases in service fee, expected return, reward-penalty intensity, and repurchase price multipliers, together with stronger fiscal subsidies for IMU and SCP, higher reputational benefits, lower water resources taxes, and moderate reductions in reserved water volumes, promote the market-based development of irrigation water-rights trading. By contrast, variations in irrigation water-use benefits, fiscal subsidies directed to IWU, and water utilisation coefficient have relatively weak effects on strategic evolution and make limited contributions to the market-based process.
- (4) Two-factor sensitivity analysis identifies synergies between subsidy rates and reputational benefits, water resources taxes and reward-penalty intensity, expected return and management-technical factors, service fee and irrigation water-use benefits, and repurchase price multiplier and irrigation water-use benefits. Overall, coordinated parameter adjustments improve system stability and accelerate strategic convergence.

Compared with existing studies (Yuan et al. 2024; Yuan et al. 2020; Yuan et al. 2023), which have mainly examined water allocation and governance from the perspectives of evolutionary conflict, bargaining-based concessions, and differential-game cooperation, this study extends the analysis to market-based irrigation water-rights trading in China by developing a four-party evolutionary game framework and clarifying the dynamic formation mechanism of market-based allocation under multi-actor interaction. The practical value of this study lies in identifying key drivers and policy combinations that support the institutional optimisation and coordinated reform in irrigation water-rights trading. Nevertheless, several limitations remain. As a theoretical evolutionary game analysis, this study relies on simplified assumptions and limited empirical validation. Future research could strengthen the framework by incorporating spatial heterogeneity, dynamic parameter adjustment, and broader empirical evidence.

Policy implications

Strengthening the mechanisms for fiscal incentives and reputational constraints

The results indicate that subsidy rates and reputational benefits jointly strengthen LCG's incentive-oriented governance capacity and promote system convergence towards a market-based equilibrium. Accordingly, LCG should adopt a more targeted policy design by linking subsidies to IMU's management performance, SCP's investment outcomes, and IWU's water-saving behaviour, while providing differentiated support for actors with different performance levels. A transparent accountability mechanism should also be established through the regular disclosure of transaction performance, subsidy use, and water-saving outcomes, with these results incorporated into government performance assessment and public oversight. In this way, fiscal incentives and reputational constraints can jointly shift LCG from direct administrative intervention towards rule-based and incentive-oriented governance, thereby improving policy implementation efficiency and enhancing the long-term stability of irrigation water-rights trading reform.

Establishing a management mechanism balancing incentives and constraints

The results suggest that service fee, subsidy rates, reward-penalty intensity, and reserved water volume have measurable effects on IMU's strategic choices. Accordingly, IMU governance should combine performance-based incentives with binding constraints. Service fee and subsidies should be linked to indicators such as water-saving performance, water-delivery stability, and operational efficiency, so that high-performing units receive greater support. A tiered tax-and-penalty mechanism should also be introduced, under which IMU with higher efficiency and stronger management performance receives moderate tax reductions, while those characterised by extensive management or high energy intensity face stricter penalties and heavier tax burdens. Reserved water volume should also be maintained within a reasonable range to avoid weakening management flexibility and trading efficiency. Through this combination of differentiated incentives, performance assessment, and tax constraints, IMU can be guided from inert management towards efficient management, thereby improving operational efficiency and the reliability of agricultural water supply.

Stabilising expected returns and strengthening performance linkages

The results indicate that SCP's strategic choices are jointly shaped by expected return, fiscal subsidies, and management efficiency. Accordingly, LCG should promote SCP participation through mechanisms that combine stable returns with performance-linked incentives. In practice, contracts should specify minimum return guarantees, revenue-sharing rules, and risk-sharing arrangements to reduce investment uncertainty, while fiscal subsidies should be provided in a timely and transparent manner. IMU should also improve management efficiency through infrastructure upgrading and smart monitoring to reduce SCP's operating risks. Economic returns should also be linked to

measurable indicators, such as water-supply reliability and water-saving performance, so that subsidies, revenue distribution, and risk compensation are tied to actual performance outcomes. In this way, SCP can be encouraged to shift from short-term participation to sustained investment, thereby providing stable financial and technical support for the market-based development of irrigation water-rights trading.

Optimising income incentives and endogenous behavioural mechanisms

The results indicate that IWU's behavioural decisions are jointly influenced by service fee levels, repurchase multipliers, water resources taxes, and irrigation water-use benefits. Accordingly, policy design should focus on strengthening IWU's water-saving incentives through differentiated charging, stable return mechanisms, and reasonable tax regulation. Specifically, service fee should be linked to actual water-saving performance, with users achieving better outcomes receiving fee reductions or subsidies, while the use of collected fees should be disclosed regularly to enhance trust. Repurchase rules and trading-revenue arrangements should also be refined by establishing dedicated funds and differentiated repurchase standards based on saved water volumes and trading participation levels, so that water-saving efforts can be translated into tangible and stable economic returns. Water resources taxes should also be kept within a reasonable range to avoid weakening users' willingness to participate. Through this combination of cost regulation, return guarantees, and transparent implementation, IWU can be guided from passive compliance towards sustained participation in water saving and water-rights trading.

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Declarations

Competing interests The authors declare no competing interests.

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